



CENTER for
EPIGENOMICS
at UC San Diego

Spatial Transcriptomics

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BNFO262, February 12, 2026



Class Overview

1. Spatial Transcriptomics

2. MERFISH (Multiplexed Error-Robust Fluorescence in situ Hybridization)

3. MERFISH Data Analyses and Tutorial



Center for Epigenomics

A leader in single-cell epigenomic technologies.

Established in 2017, the Center is strategically built, equipped, and staffed to advance transformative approaches to probing the epigenome at unprecedented **scale** and **resolution**.

We accomplish this mission through continuous **technology innovation**, **open collaboration**, and **rapid data sharing**.





Center for Epigenomics: Scalable, reproducible, accurate

Spatial Genomics and Epigenomics (Leader: Dr. Nathan Zeng)

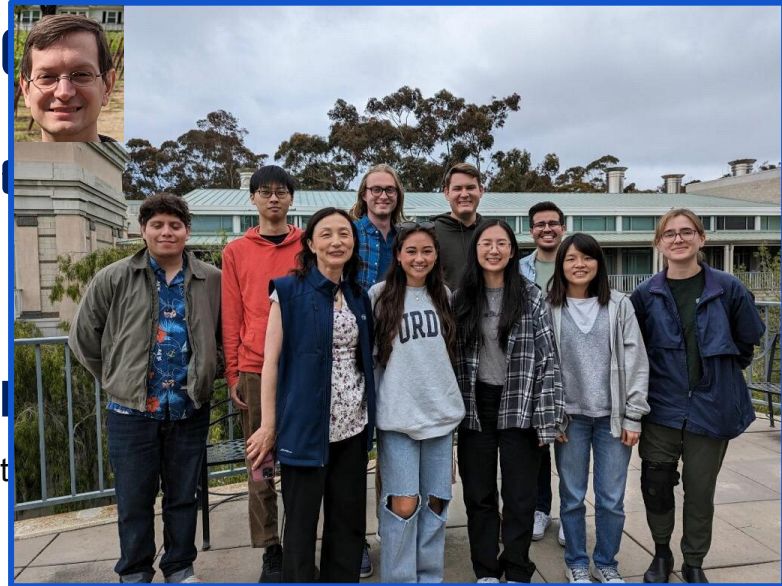
- High throughput spatial transcriptomics and spatial genomics
- Production pipelines MERFISH+, MERFISH and Xenium

Single Cell Epigenomics (Leader: Dr. Nathan Zeng)

- Automated protocols for the next generation multi-modal single-cell transcriptomics
- Multiple single-cell multiomics technologies unique to the Center for Epigenomics

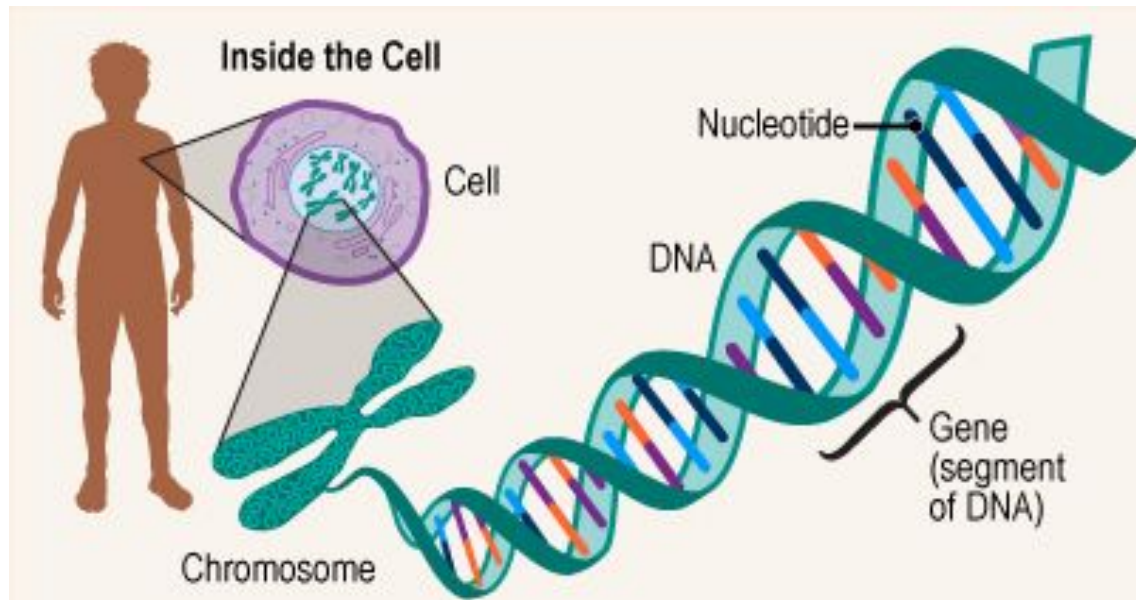
High-Throughput Epigenomics (Leader: Dr. Allen Wang)

- Assay samples at scale, cost-effective and efficient, on human biopsies and clinical samples.
- Fully integrated sample tracking, QC infrastructure, and data processing pipelines.





Gene Expression Determines Cellular Composition and Function



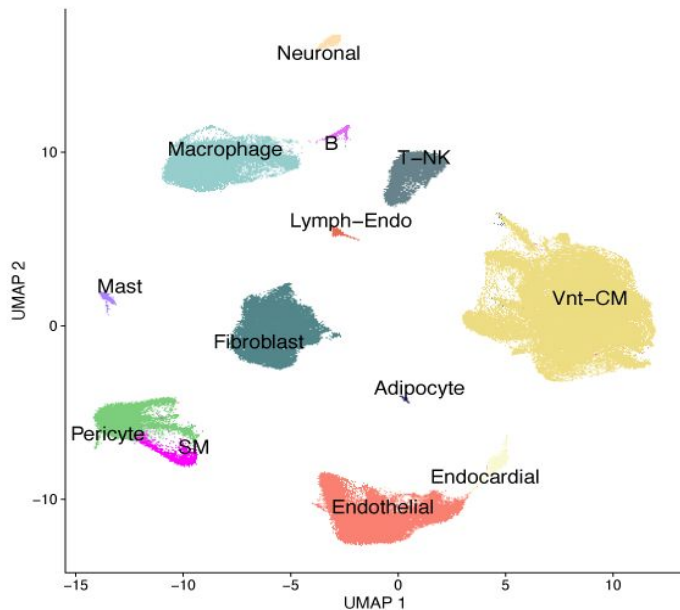
cellular composition & function





RNA Sequencing Does Not Give us Spatial Location of Cells: a big question

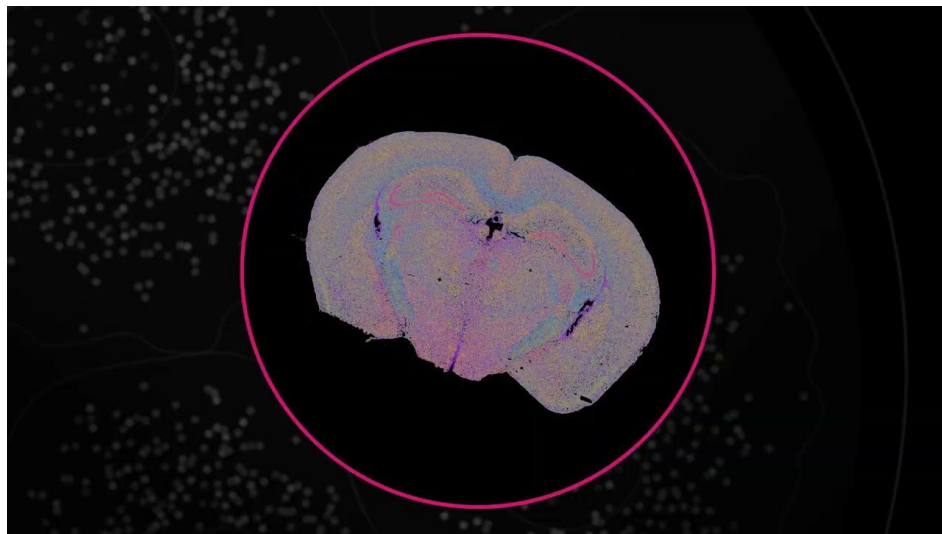
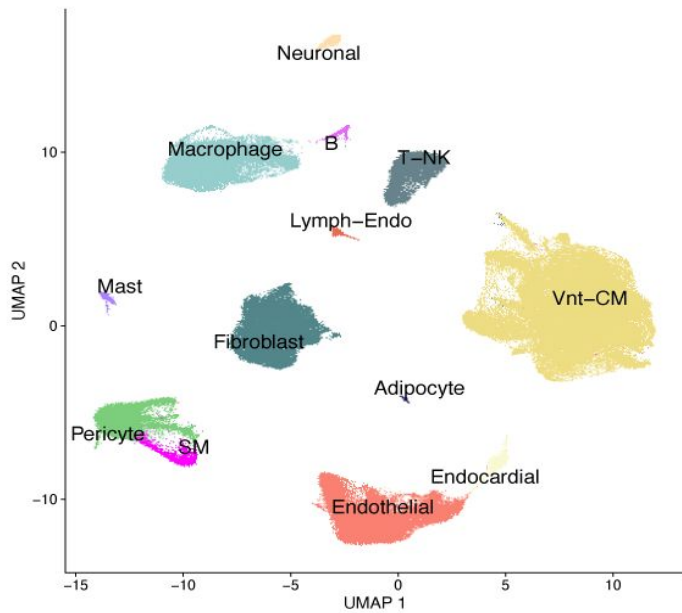
RNA Sequencing UMAP Projection





RNA Sequencing Does Not Give us Spatial Location of Cells: a big question

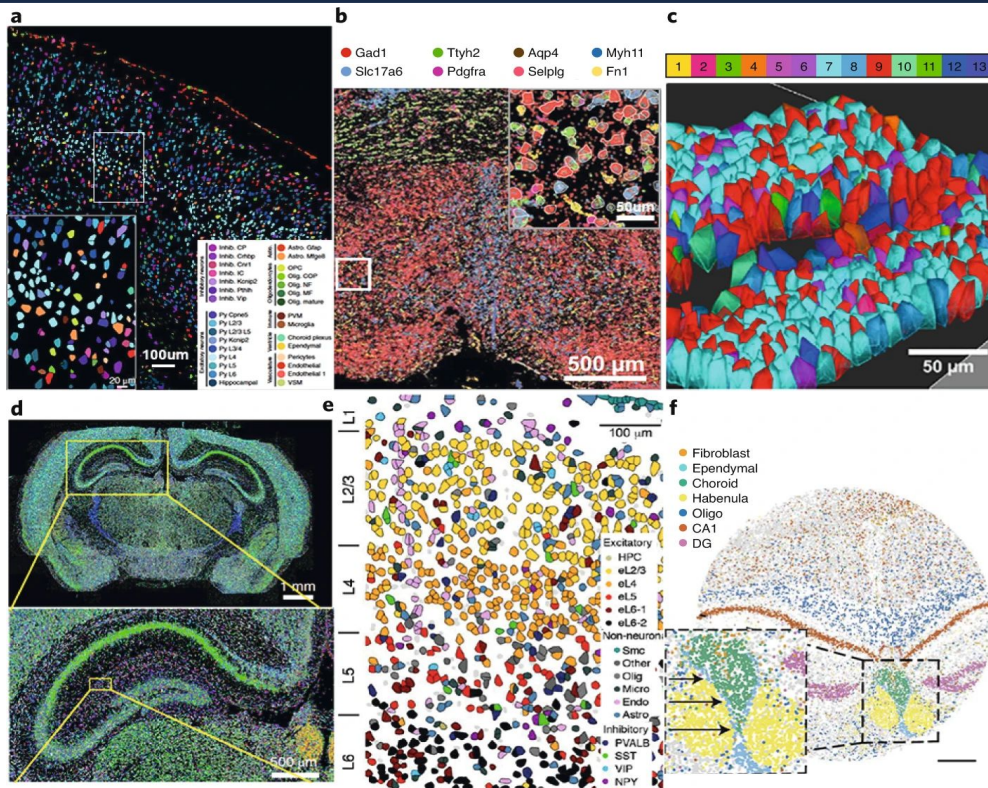
RNA Sequencing UMAP Projection



Vizgen.com

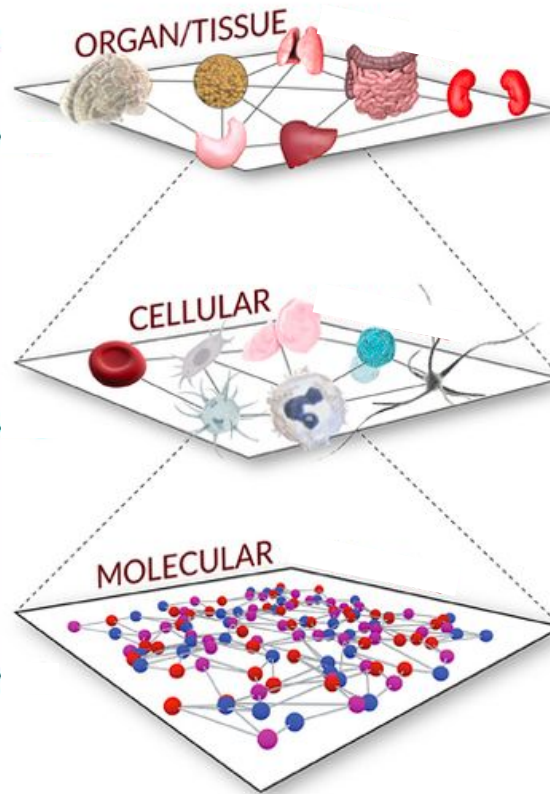
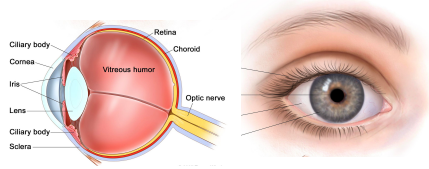


Method of the Year 2020: Spatially-resolved Transcriptomics



Close, J.L., Long, B.R. & Zeng, H. *Nat Methods* (2021).
<https://doi.org/10.1038/s41592-020-01040-z>

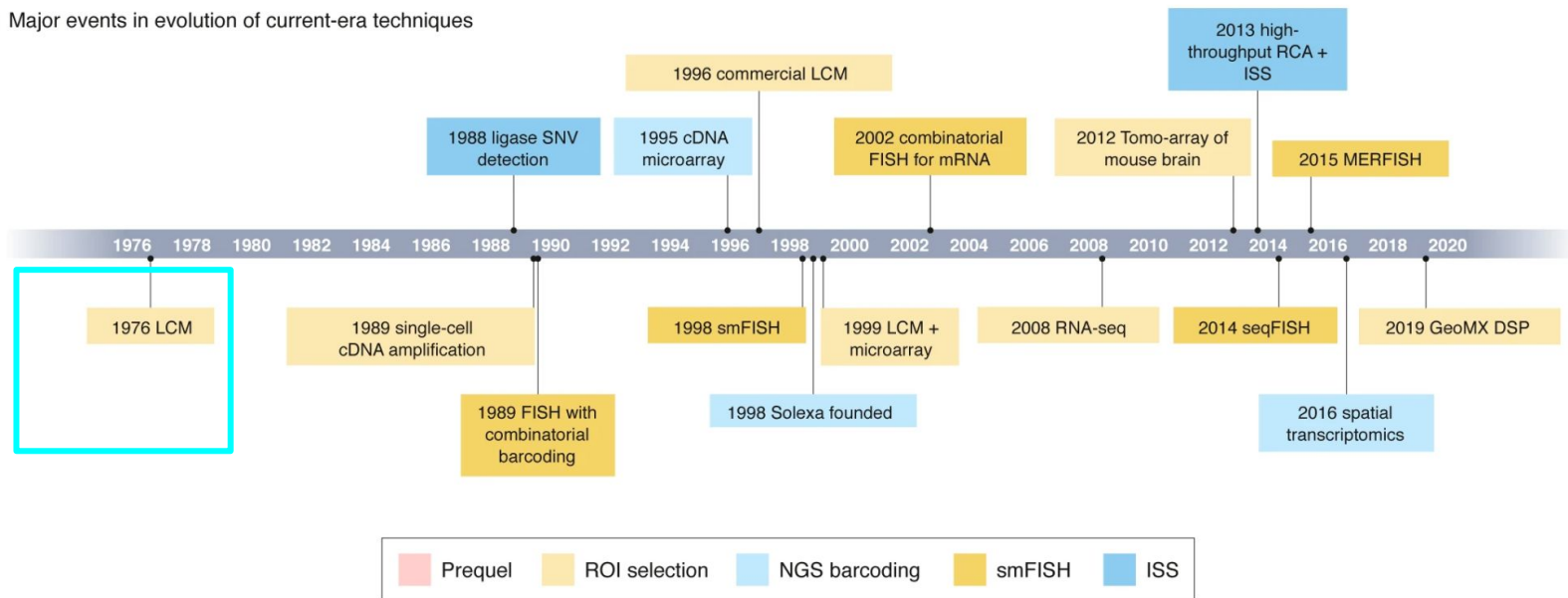
Biology at Molecular Scale , Cellular Scale , Tissue Scale





Timeline of Spatial Transcriptomics

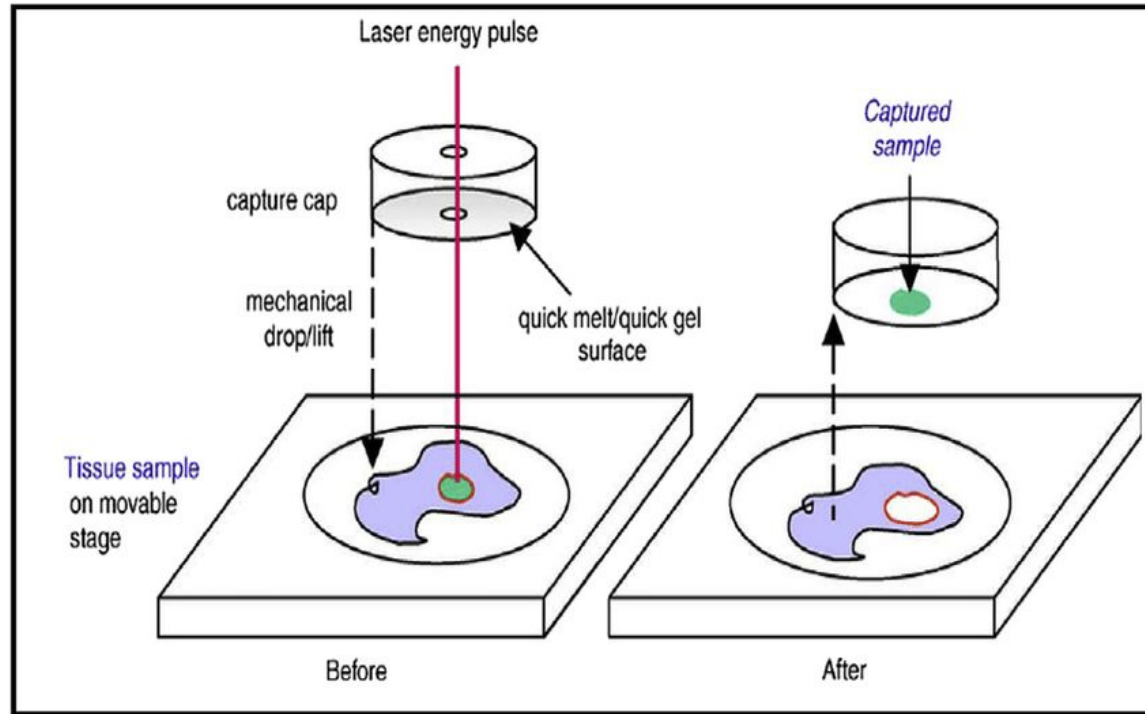
c Major events in evolution of current-era techniques



Moses, L., Pachter, L. *Nat Methods* (2022).
<https://doi.org/10.1038/s41592-022-01409-2>



Laser Capture Microscope: bridging histology and molecular profiling

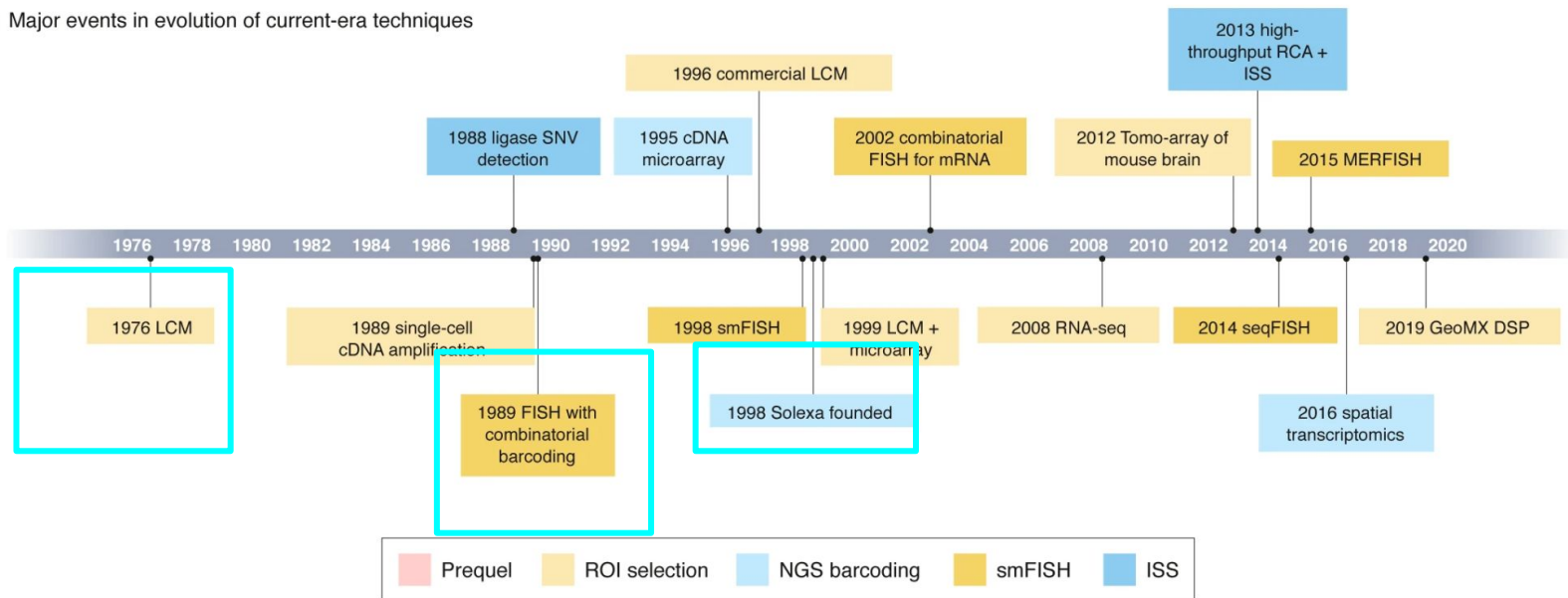


**low throughput,
hypothesis driven**



Timeline of Spatial Transcriptomics

c Major events in evolution of current-era techniques

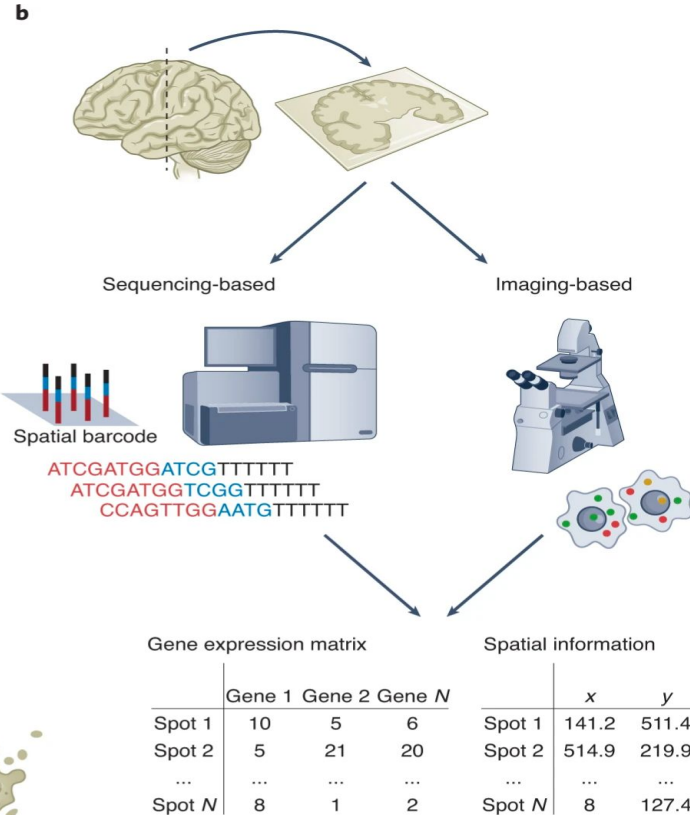


Moses, L., Pachter, L. *Nat Methods* (2022).
<https://doi.org/10.1038/s41592-022-01409-2>

Approaches for Spatial Genomics

‘sequencing-based ST’ (sST): capture RNA locally from intact tissue sections on a pixelated, DNA-barcoded surface and read out their gene identities using next-generation sequencing.

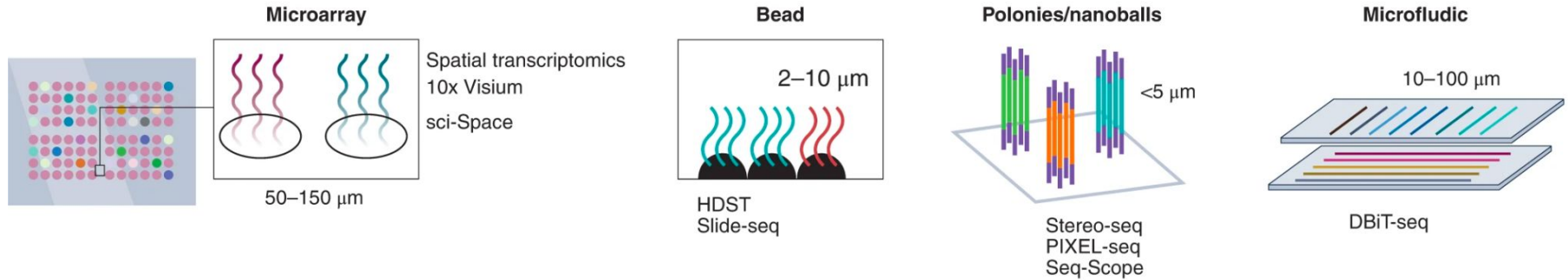
imaging-based ST’ (iST)
working on microscopy techniques
simultaneously detecting the presence of
many mRNA transcripts within tissues using
fluorescence in situ hybridization (FISH) or
direct in situ sequencing



Tian, L., Chen, F. & Macosko, E.Z. The expanding vistas of spatial transcriptomics. *Nat Biotechnol* (2022). <https://doi.org/10.1038/s41587-022-01448-2>

NG-Sequencing-based Spatial Transcriptomics: overview

1. Spatially indexing surface with DNA-oligo barcodes



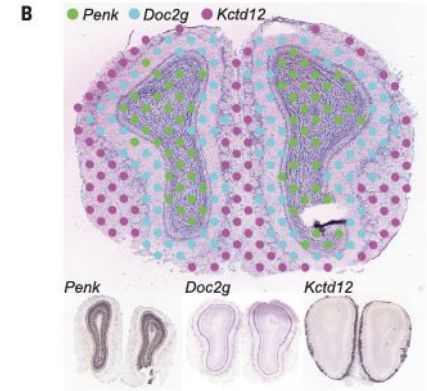
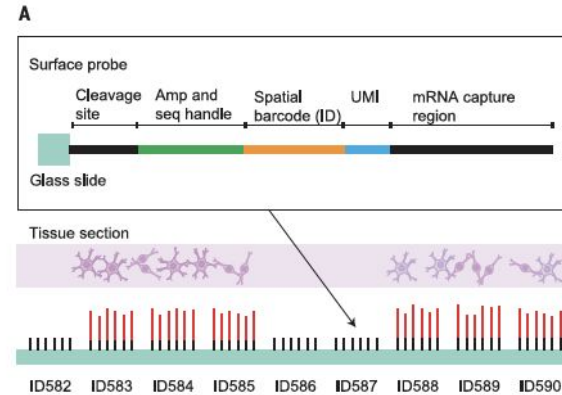
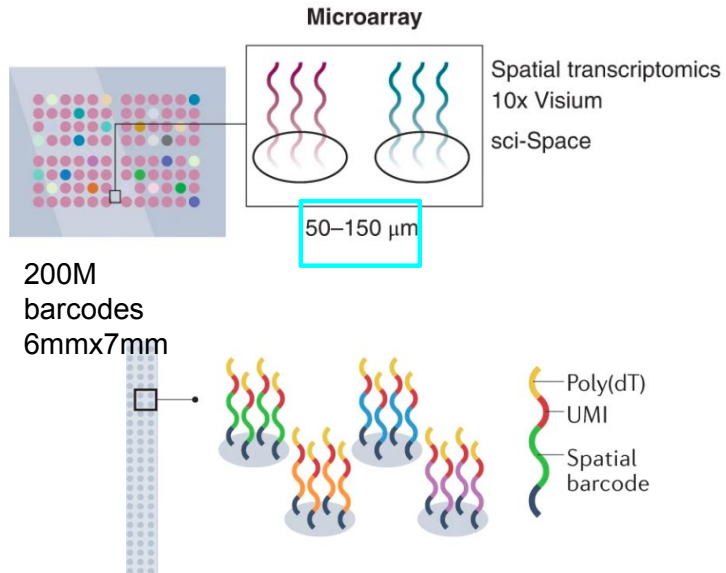
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<https://doi.org/10.1038/s41587-022-01448-2>

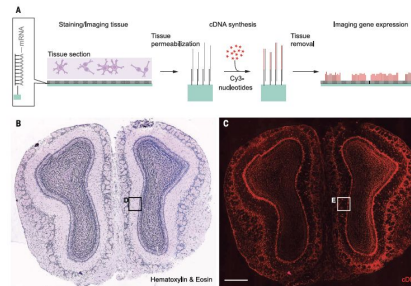
<https://doi.org/10.1038/s41576-022-00515-3>

NG-Sequencing-based Spatial Transcriptomics: ST

1. Spatially indexing surface with DNA-oligo barcodes: Spatial Transcriptomics (ST)



15

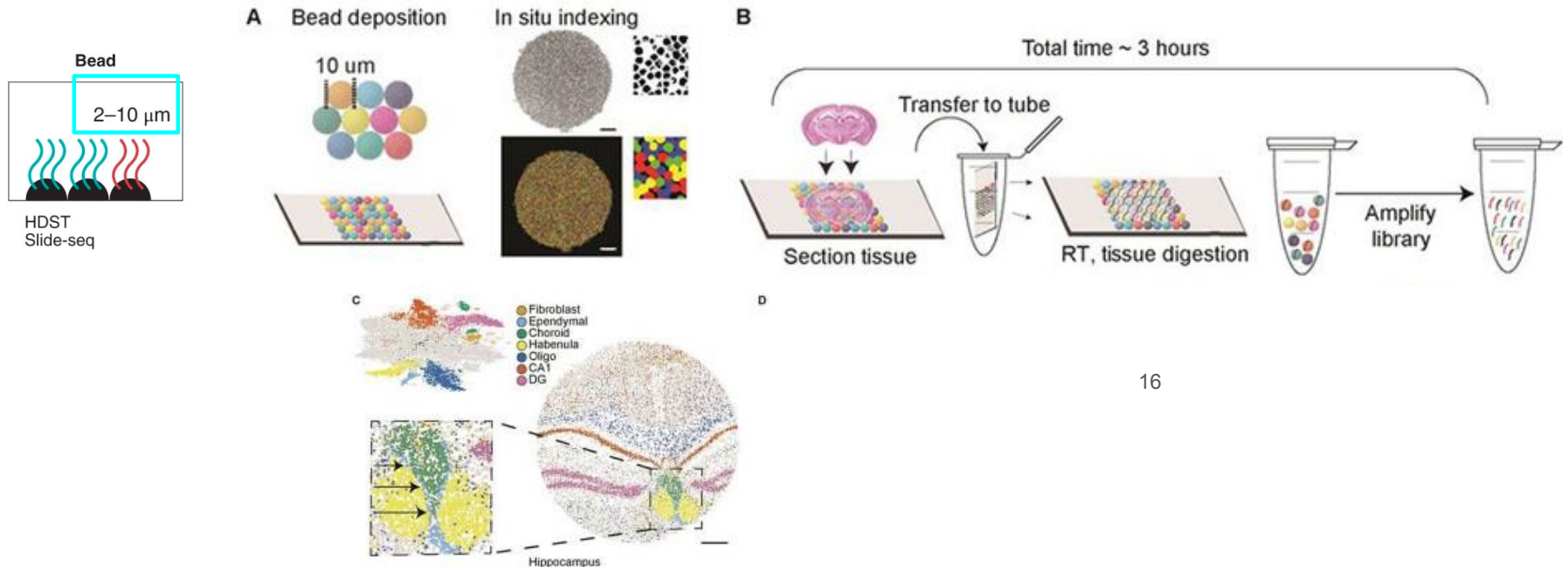


Stahl et al., JONAS FRISÉN, Science 2016

DOI: [10.1126/science.aaf2403](https://doi.org/10.1126/science.aaf2403)

NG-Sequencing-based Spatial Transcriptomics: Slide-seq

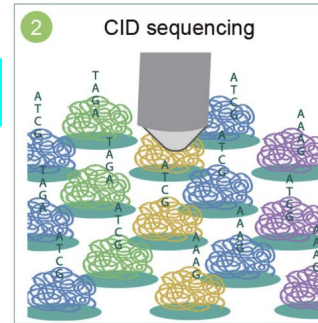
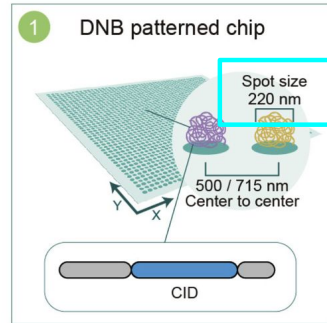
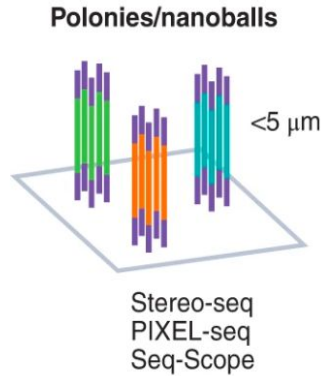
1. Spatially indexing surface with DNA-oligo barcodes: Slide-seq



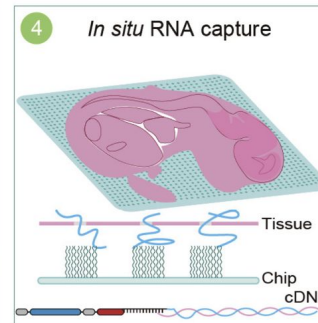
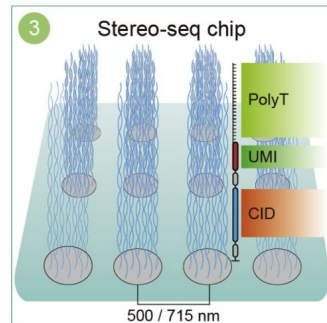
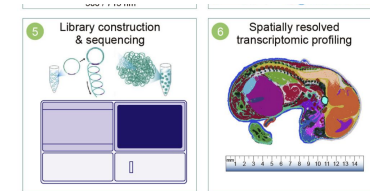
16

NG-Sequencing-based Spatial Transcriptomics: Stereo-seq

2. Spatially indexing surface with DNA-oligo barcodes: Stereo-seq



barcode coordinate identity (CID)

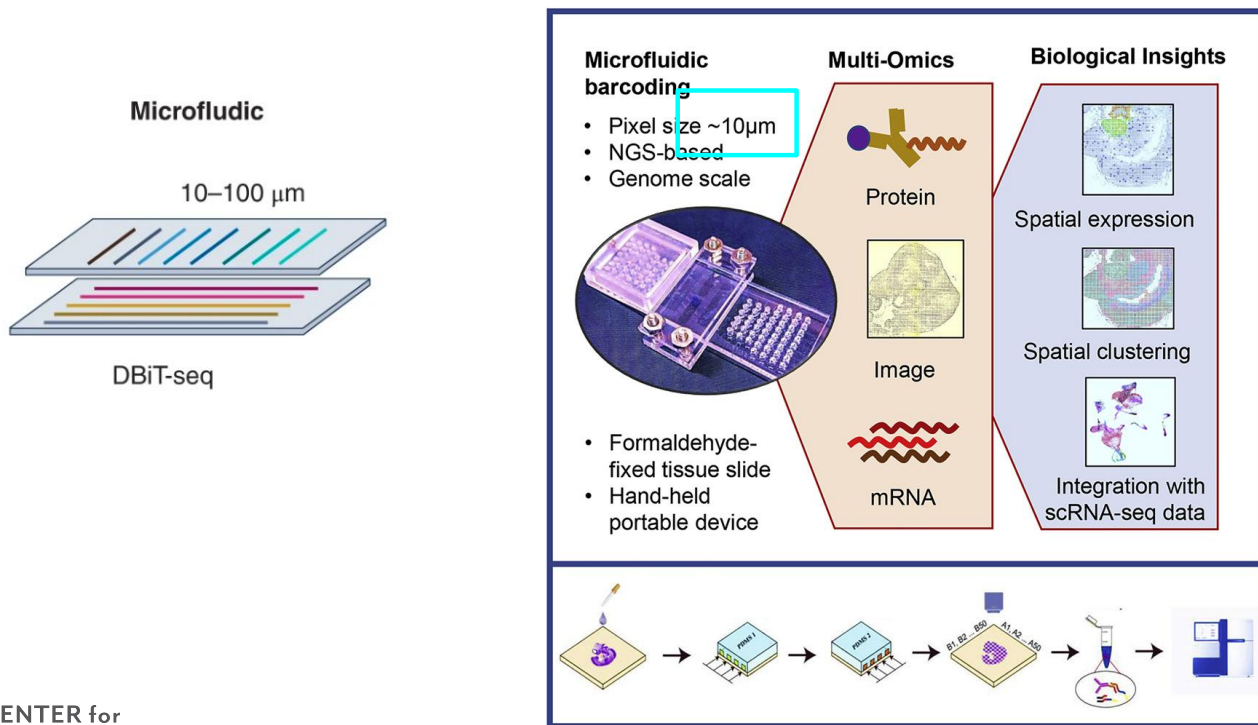


17

6mmx10mm

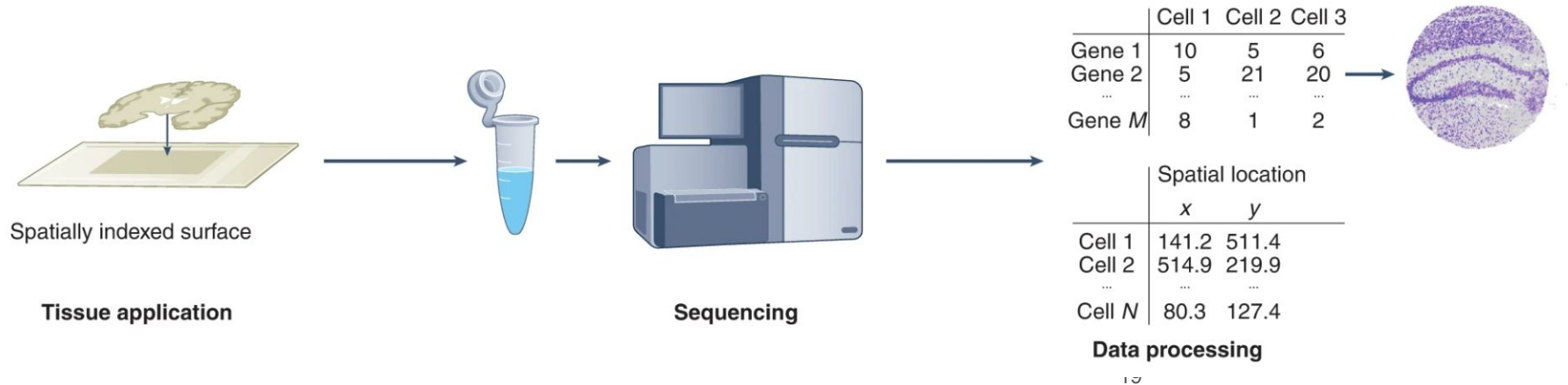
NG-Sequencing-based Spatial Transcriptomics: DBiT-seq

3. Spatially indexing surface with DNA-oligo barcodes: DBiT-seq Multi-modal (RNA, DNA, protein)



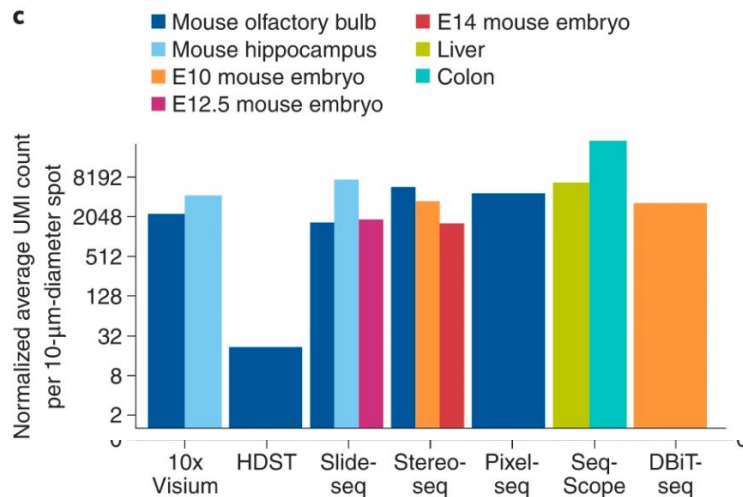
NG-Sequencing-based Spatial Transcriptomics: Summary

Generation of NGS-transcriptome paired with spatial location

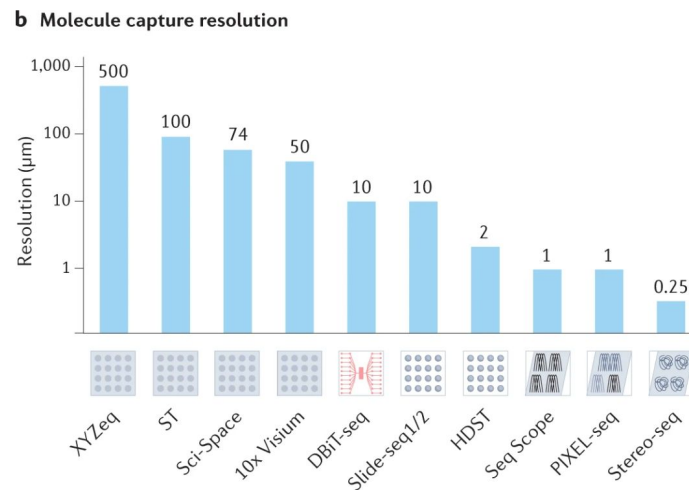


Choices of NG-Sequencing-based Spatial Transcriptomics

mRNA Capture Sensitivity

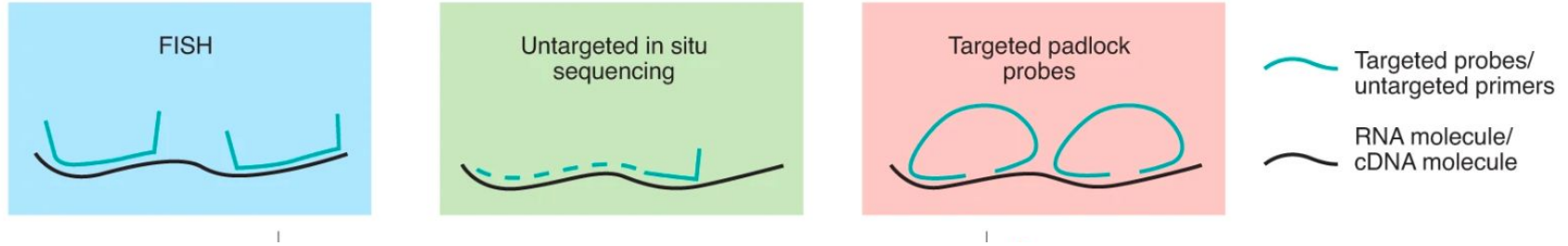


Spatial Resolution:



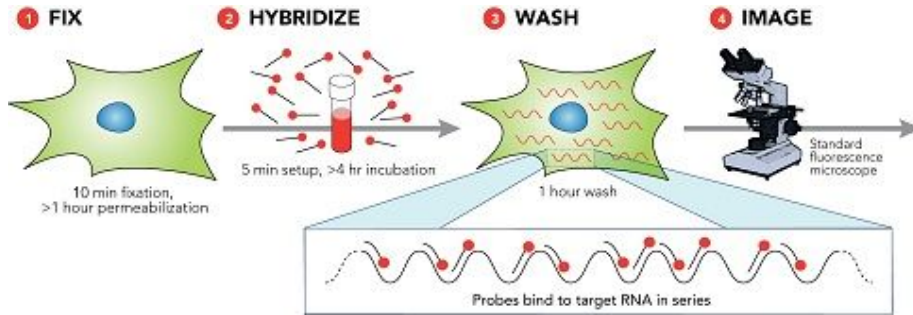
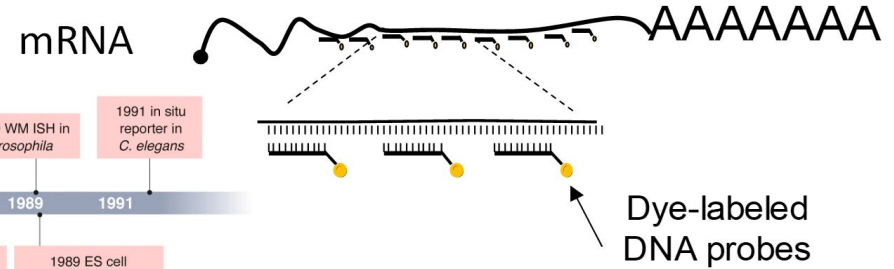
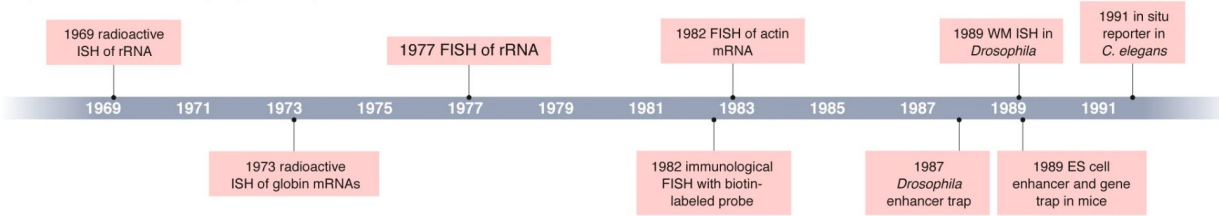
Moffitt et al., Nat Rev Gen 2022 <https://doi.org/10.1038/s41576-022-00515-3>

Imaging-based Spatial Transcriptomics



What is Fluorescence *in situ* Hybridization (FISH)?

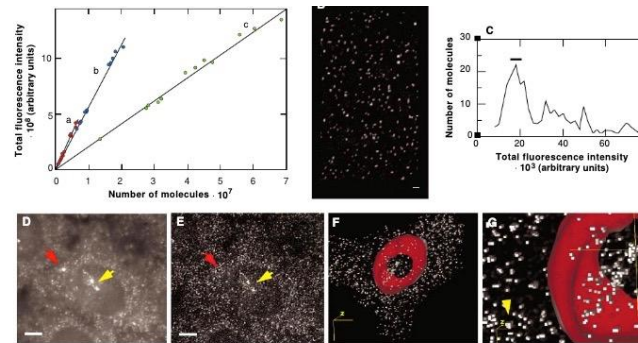
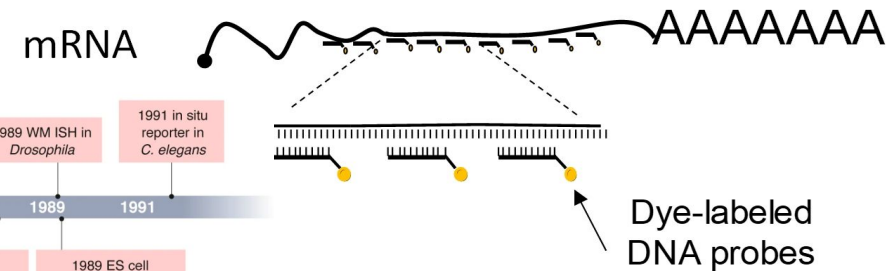
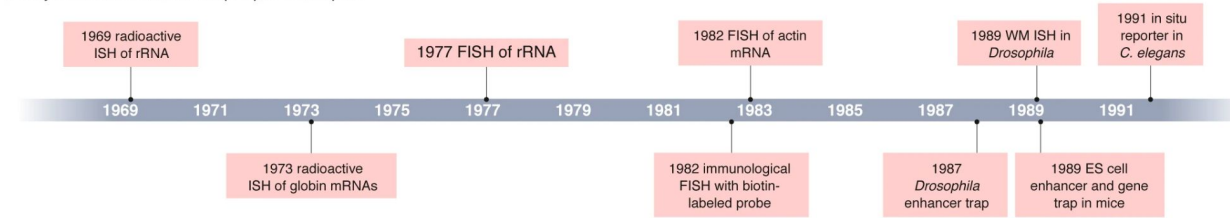
a Major events in evolution of prequel techniques





What is Fluorescence *in situ* Hybridization (FISH) single-molecule (sm)FISH

a Major events in evolution of prequel techniques

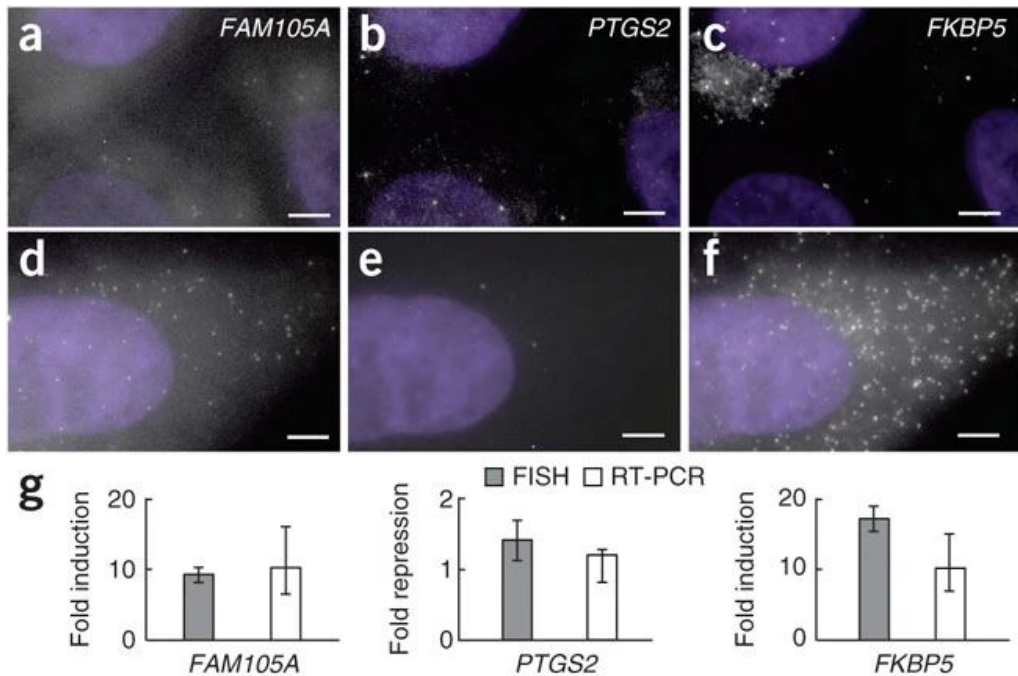


Femino et al.,... Singer, Science, 1998

Fig. 1. Methodology for detection of single RNA molecules by quantitative FISH and digital imaging microscopy (13). Probes used are described in (14). **(A)** The amount of light emitted per probe was calibrated by measuring the TFI from a known number of probes in an imaged volume (15). The TFI was plotted against the number of fluorochrome molecules (five per probe) to generate the regression curve. The slope is equal to the TFI per fluorochrome molecule (curve a = CY3, curve b = CY5, curve c = FITC). A normalized TFI per probe was used to calculate the number of probes hybridized at discrete point sources of light in a restored image (15). **(B)** The accuracy of the technique was tested by optically sectioning and restoring immobilized CY3-labeled probes adsorbed to glass cover slips. **(C)** The most frequent occurrences of TFI values per CY3 probe plotted as a histogram are comparable to the normalized value (solid bar) from the solution experiments (16). **(D)** One optical plane of an NRK cell after in situ hybridization (17) with calibrated CY3 probes to β -actin mRNA shows a number of bright foci superimposed on a diffuse background arising from out-of-focus light. Red arrow, bead = 0.092 μ m; yellow arrow, bead = 0.092 μ m. **(E)** The 3D-reconstructed image of the same area as shown in (D) shows the same bright foci superimposed on a diffuse background arising from out-of-focus light. Red arrow, bead = 0.092 μ m; yellow arrow, bead = 0.092 μ m.



Simultaneous Imaging of Three Different mRNAs in Mammalian Cells

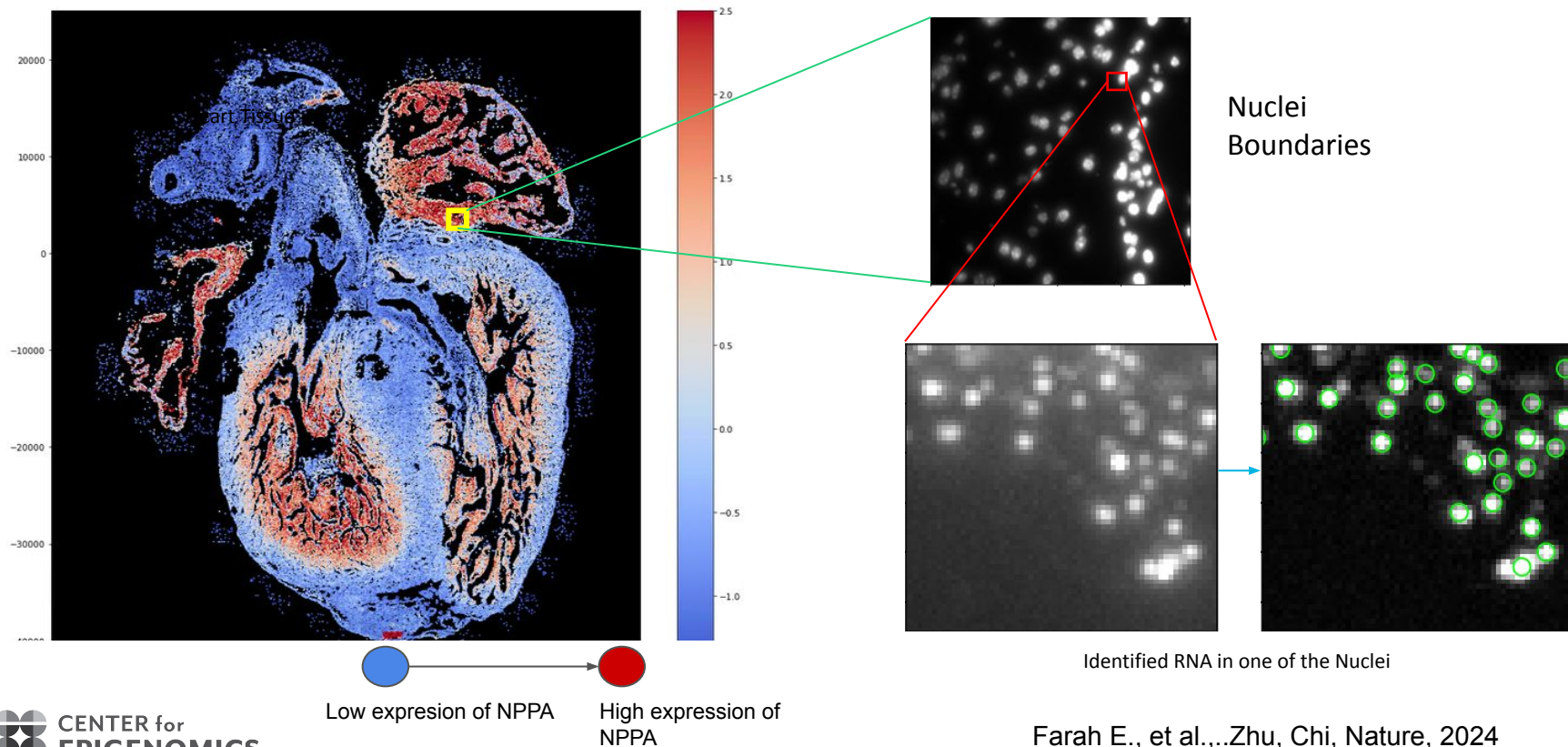


Raj et al... Tyagi, 2008
<https://doi.org/10.1038/nmeth.1253>





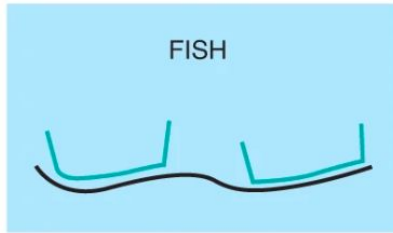
Imaging single mRNAs in human tissues



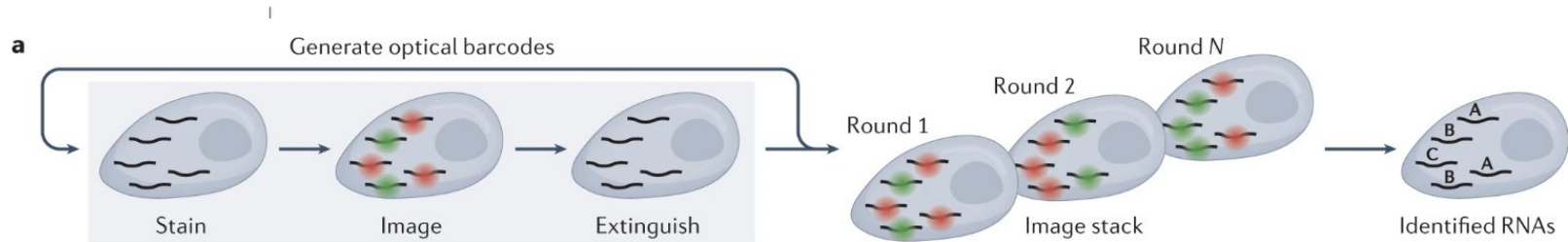
Farah E., et al., Zhu, Chi, Nature, 2024



Multiplex smFISH Imaging



- Oligonucleotide synthesis
- Fluorescence microscopy
- Single-cell transcriptomics

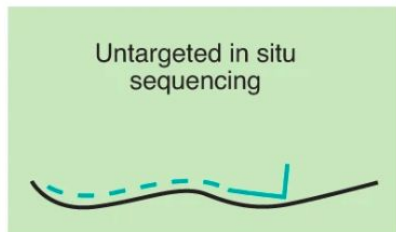


Moffitt et al., 2022, <https://doi.org/10.1038/s41576-022-00515-3>

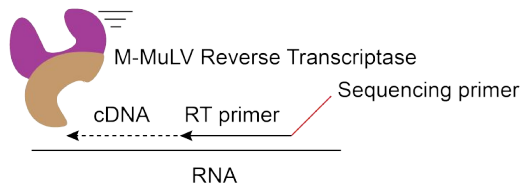


Imaging-based Spatial Transcriptomics-in situ Sequencing: FISSEQ

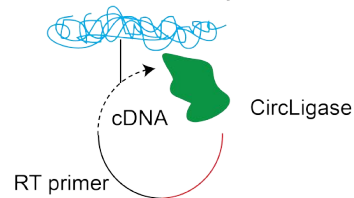
Ligation-based imaging sequencing: FISSEQ



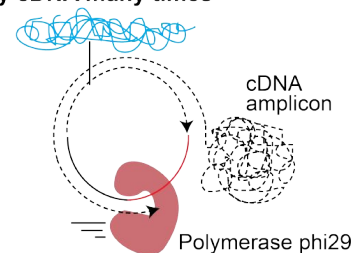
Reverse Transcription (RT)



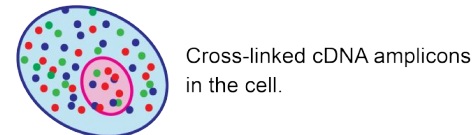
Circularize cDNA template



Copy cDNA many times

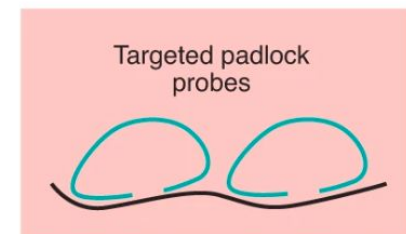


Sequence the cDNA amplicons in the cell



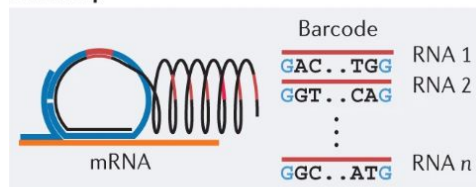


Imaging-based Spatial Transcriptomics: STARmap

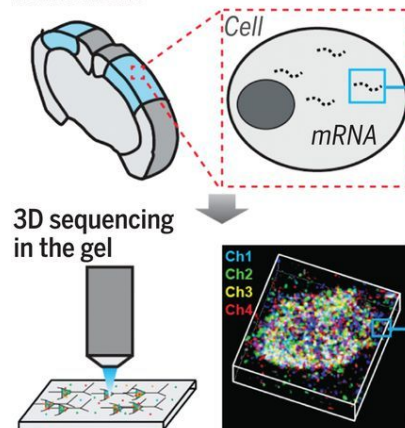


STARmap

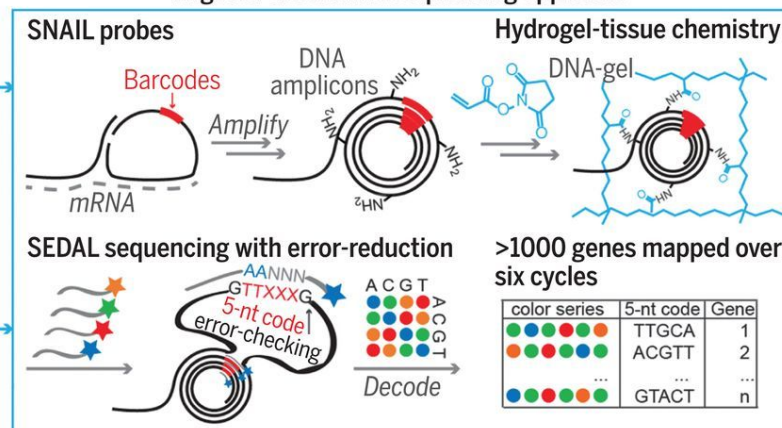
STARmap



Intact tissue



Targeted in situ RNA sequencing approach



Wang et al.,, Delsserth, Science, 2018. DOI: [10.1126/science.aat5691](https://doi.org/10.1126/science.aat5691)

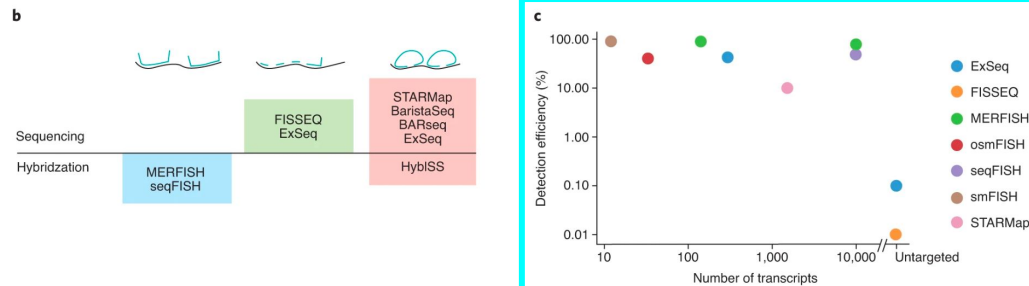
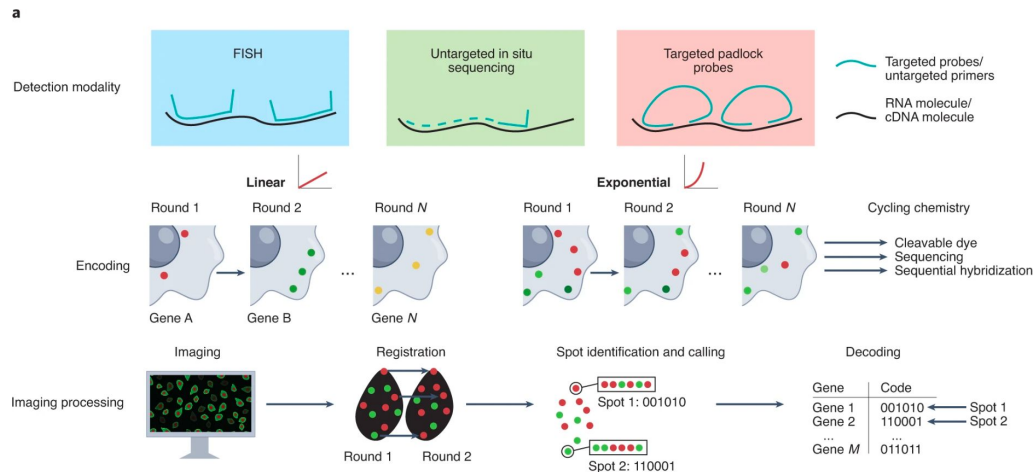


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Summary: Choices of Imaging-based Spatial Transcriptomics

- Imaging processing
- Detection efficiency
- Specificity

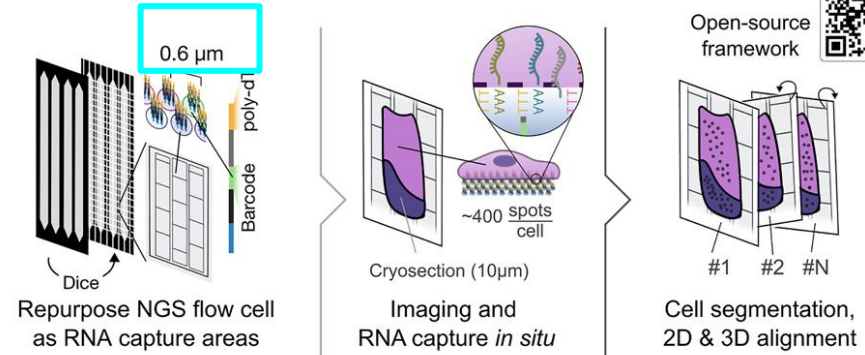


<https://doi.org/10.1038/s41587-022-01448-2>

Emerging Spatial Transcriptomics 2026

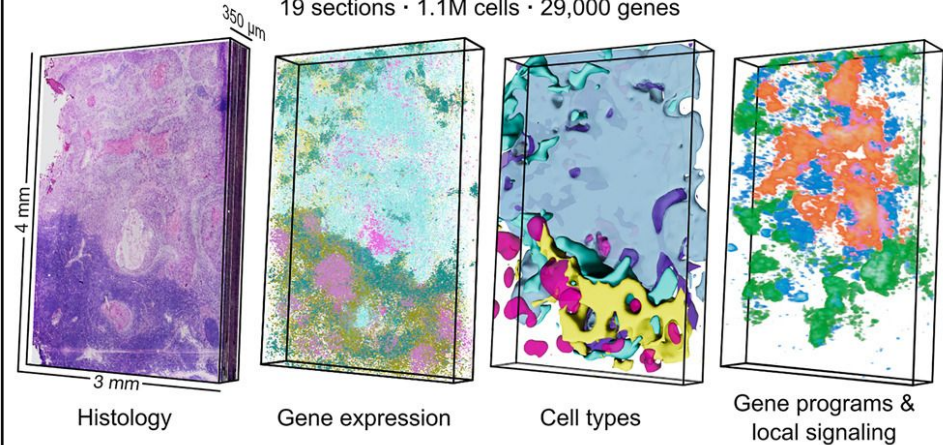
- Easy to use
- High resolution
- Cost-efficient
- 3D-scalable

Open-ST: Do-it-yourself, subcellular, spatial transcriptomics in 2D or 3D



3D virtual tissue block: human metastatic lymph node

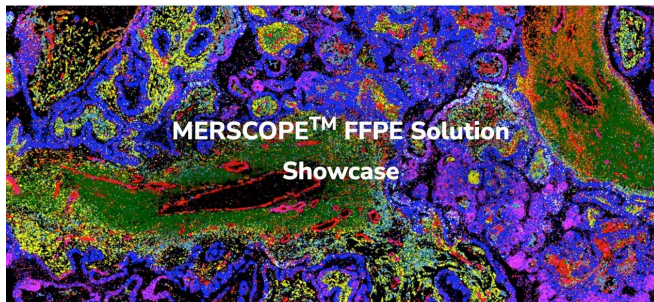
19 sections • 1.1M cells • 29,000 genes





Commercialization of Spatial Genomics 2023

vizgen



Unlock High-Quality FFPE Spatial Genomic Analysis with MERSCOPE

nanoString

PRODUCTS | RESEARCH FOCUS | RESOURCES | SUPPORT | COMPANY

CosMx™ SMI FFPE Dataset

[COSMX SMI OVERVIEW](#) [TECHNOLOGY ACCESS PROGRAM](#) [GRANT SUPPORT](#)

Products | CosMx Spatial Molecular Imager | CosMx SMI FFPE Dataset

What will you discover?

[CosMx Spatial Molecular Imager](#) (taking orders now) represents the next evolution in spatial biology. CosMx SMI is a true multiomic, [single-cell solution](#) that enables researchers to investigate morphologically intact tissues at unprecedented

10x GENOMICS

Products | Area of Interest | Resources | Support | Company | Careers | Search

Xenium In Situ Analysis Technology

High resolution, high-target analysis

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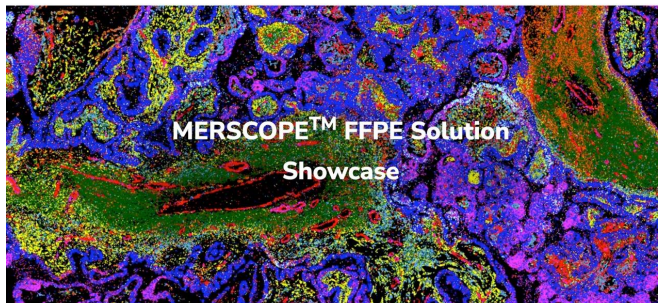
Key Features | How it works | Applications | Contact us

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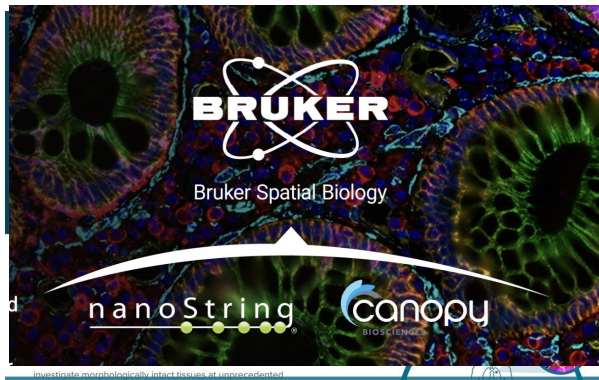


Commercialization of Spatial Genomics 2026

vizgen



Unlock High-Quality FFPE Spatial Genomic Analysis with MERSCOPE



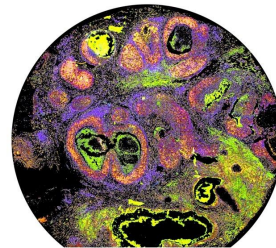
10x genomics

Products Area of Interest Resources Support Company Careers Search

Xenium In Situ Analysis Technology

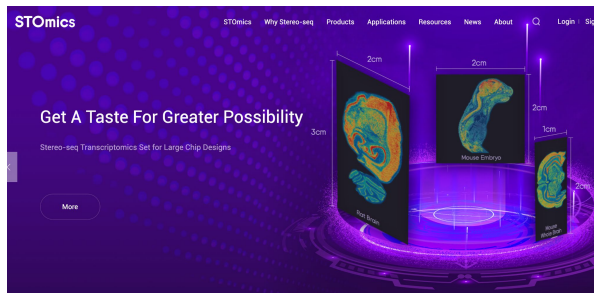
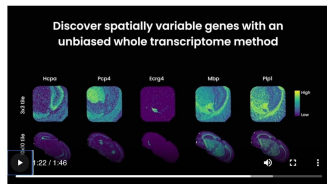
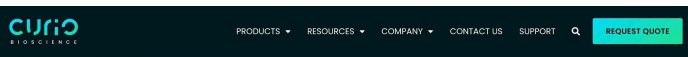
High resolution, high-target analysis

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Key Features How it works Applications Contact us

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MERFISH Established as Leading Technology

MERFISH has already been recognized as a robust technology for adoption



First published in 2015 from
Xiaowei Zhuang's lab

12 peer reviewed publications

PLUS TWO IN PREPRINT



nature|methods

nature
COMMUNICATIONS

PNAS

Proceedings of the
National Academy of Sciences
of the United States of America

molecular
systems
biology

Methods in
ENZYMOLGY

SCIENTIFIC
REPORTS



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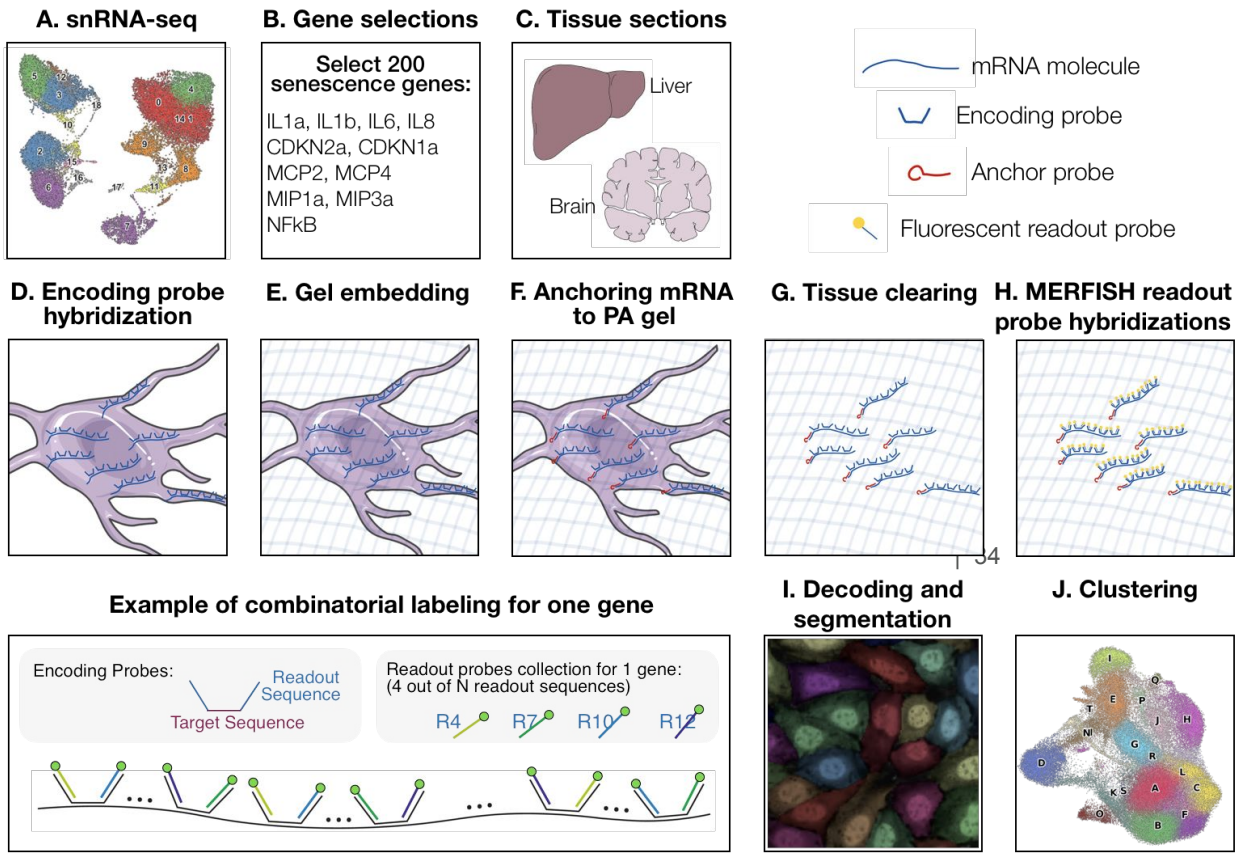


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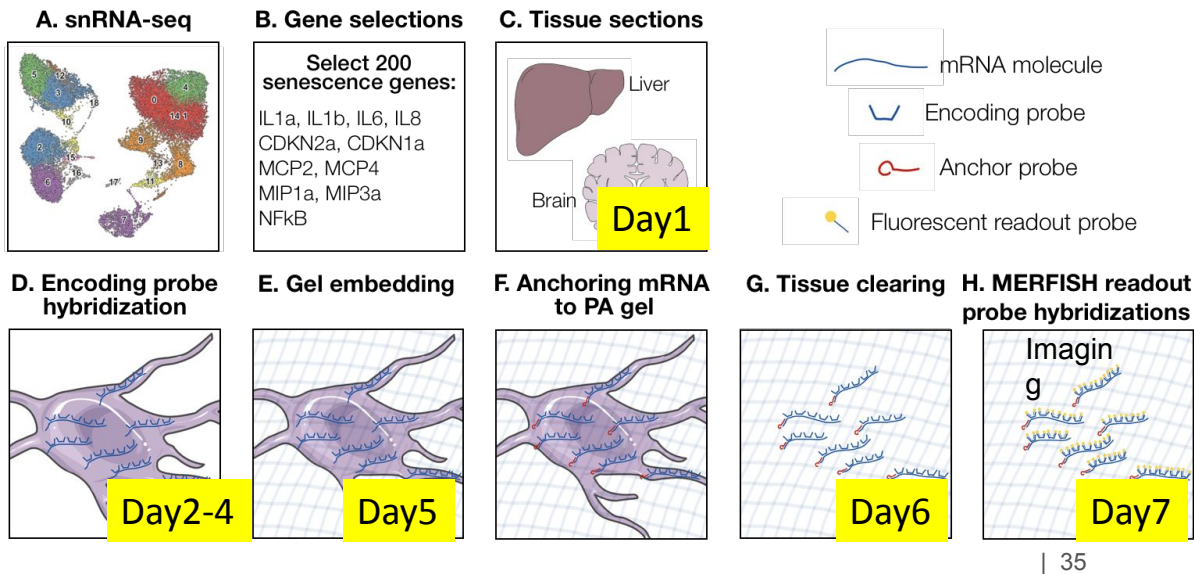


MERFISH Workflow and Tissue Clearing





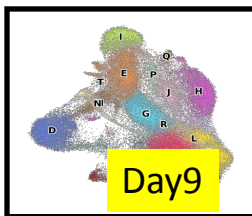
Tissue MERFISH Experiment Timeline



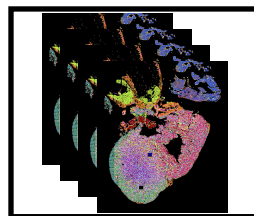
Decoding and
segmentating



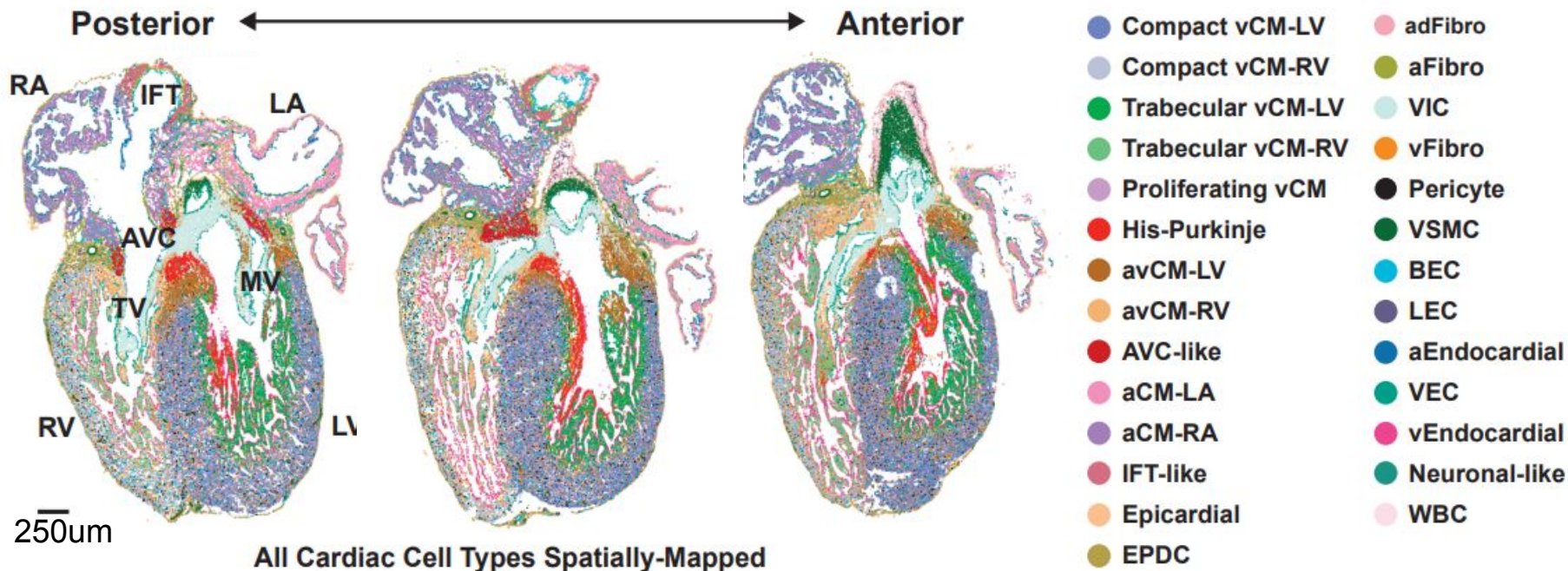
Clustering



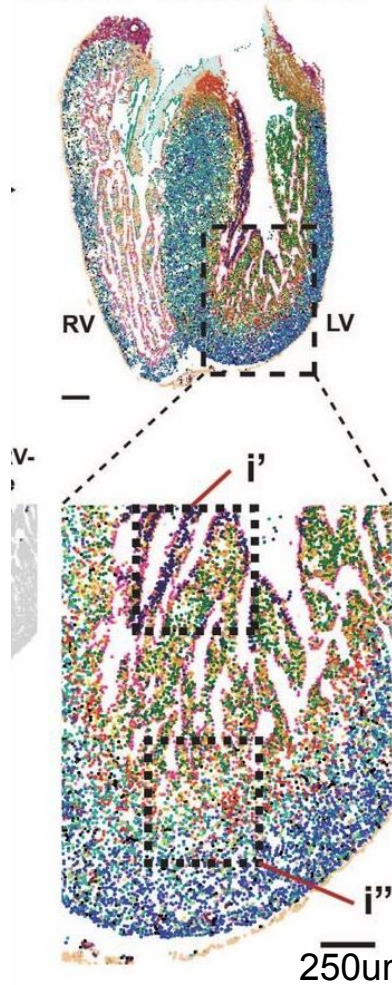
Spatial mapping



MERFISH Reveals Spatial Distribution of 27 Cell Types, 260,000 Cells in Developing Human Heart

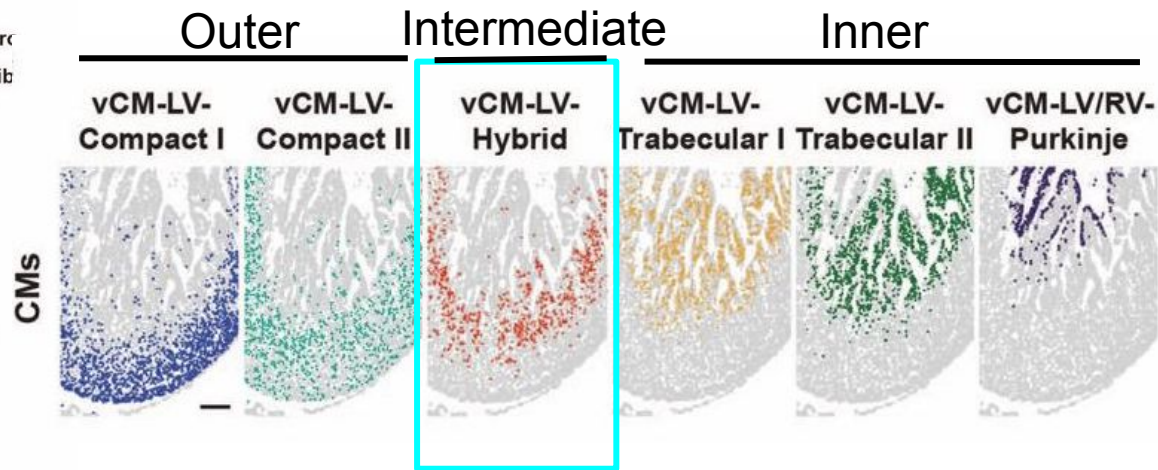


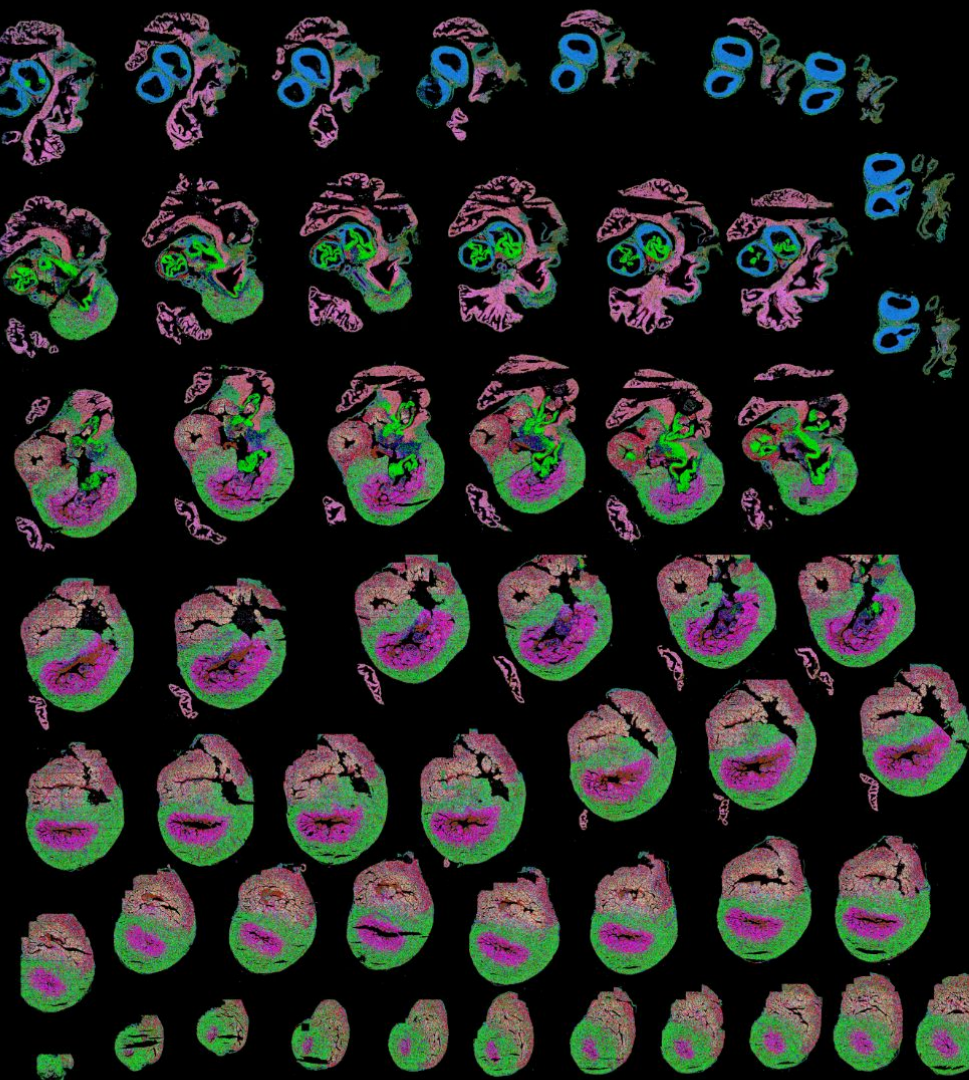
Spatially Mapped Identified Ventricular Cells



- vCM-LV-Hybrid
- vCM-LV-Compact I
- vCM-LV-Compact II
- vCM-LV-Trabecular I
- vCM-LV-Trabecular II
- vCM-RV-Hybrid
- vCM-RV-Compact
- vCM-RV-Trabecular
- vCM-RV-Prolif.
- vCM-LV/RV-Purkinje
- vCM-IVS-His
- vCM-LV-AV
- vCM-RV-AV
- Compact vFibro
- Trabecular vFib
- Prolif. vFibro
- EPDC
- Epicardial
- VIC
- Pericyte
- VSMC
- vEndocardial
- VEC
- LEC
- BEC
- WBC
- Neuronal

MERFISH Identified Novel Cell Types and Spatially Organized Complex Laminar Structure: 27 Cell Types, 260,000 Cells

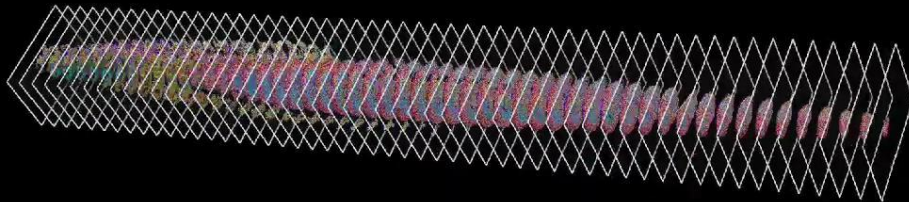




3D Reconstruction of One Human Heart 3.5 Million Cells

- | | |
|---------------------|-------------|
| ● BEC | ● Trabecula |
| ● Compact vCM-LV | ● VIC |
| ● Compact vCM-RV | ● VSMC |
| ● EPDC | ● aCM-LA |
| ● Endocardial | ● aCM-RA |
| ● His-Purkinje | ● aFibro |
| ● LEC | ● iftFibro |
| ● Neuronal | ● ncCM-IFT |
| ● Pericyte | ● vCM-LV-A |
| ● Proliferating vCM | ● vEndocar |
| ● Trabecular vCM-LV | ● vFibro |

MERFISH+

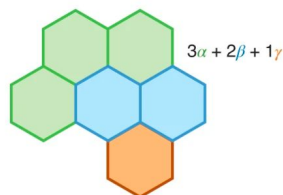


**3D Human Heart:
3.5 Million Cells**



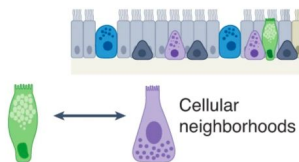
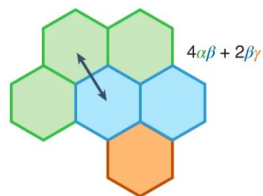
Three Primary Classes of Biological Questions Addressed by Spatial Transcriptomics

Cell composition



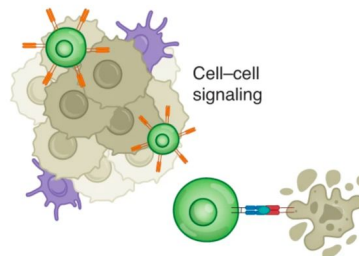
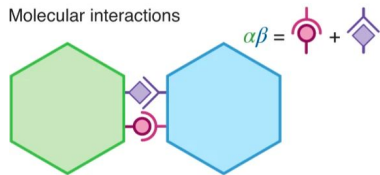
Tissue atlases

Cell-cell interactions



S_i

Molecular interactions



Tian, L., et al., *Nat Biotechnol* (2022).
<https://doi.org/10.1038/s41587-022-01448-2>

Applying MERFISH to Your Projects

Dr. Colin Kern

Outline of a MERFISH project

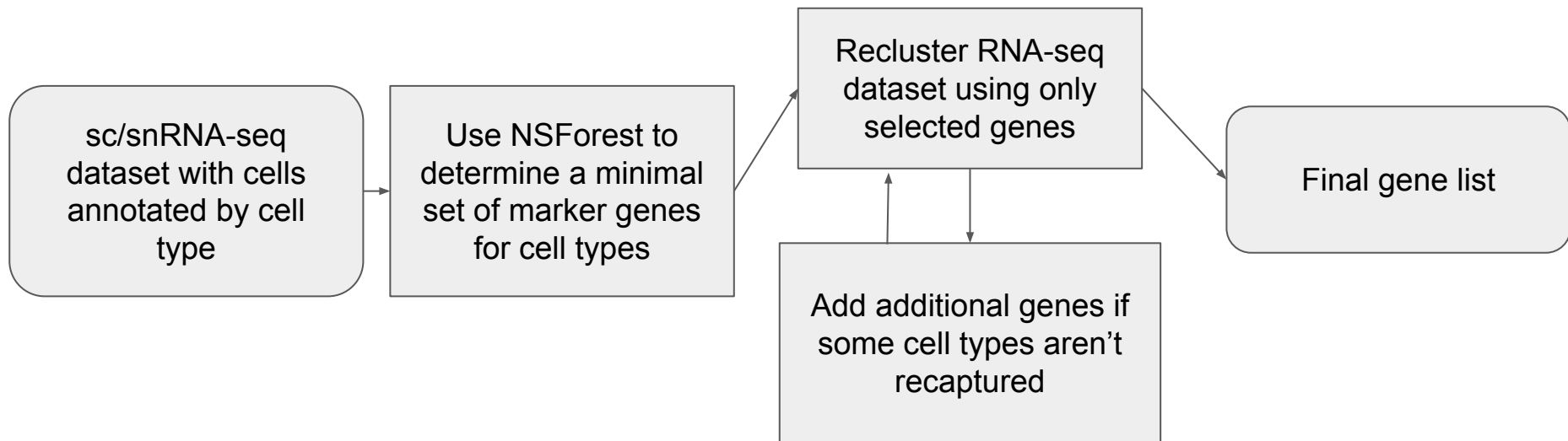
- Gene selection
- Codebook and probe design
- Sample preparation
- Imaging
- Image processing
- Data analysis

MERFISH Gene Selection

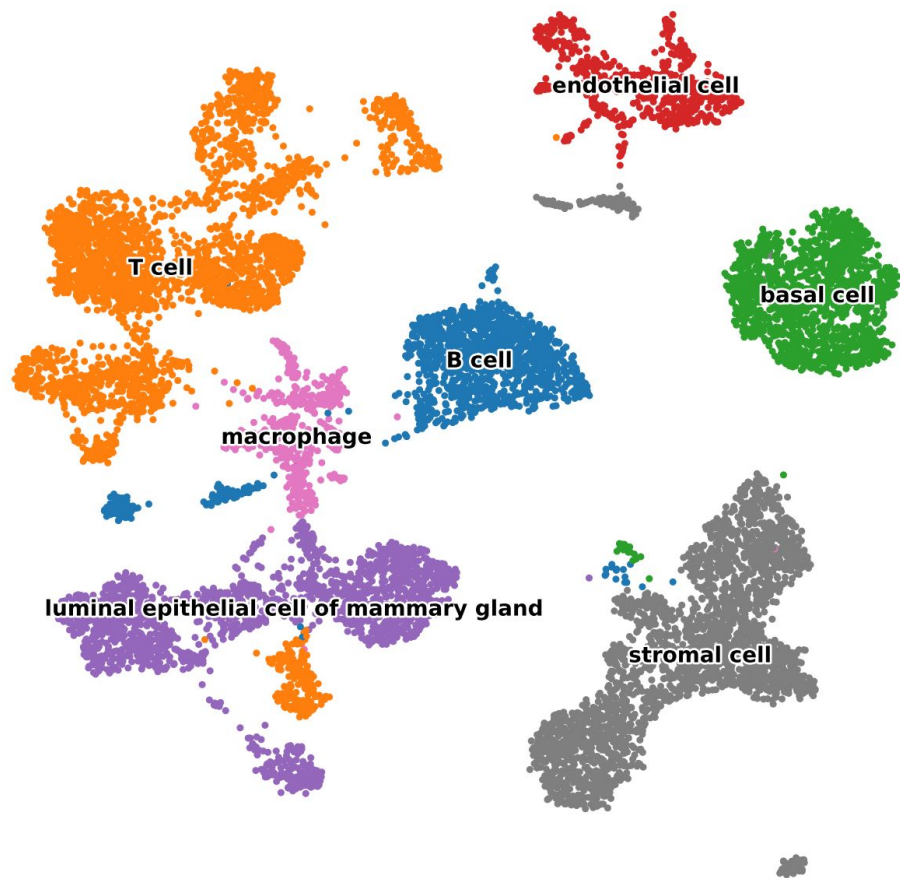
Goal of gene selection

- Cell type identification
 - scRNA-seq data for reference
 - Canonical markers aren't necessarily the best candidates
- Genes of interest
 - Differential genes in treatment
 - Markers of cell state, e.g. senescence
- Current state of MERFISH technology
 - Vizgen: Up to 1000 genes

MERFISH gene selection procedure



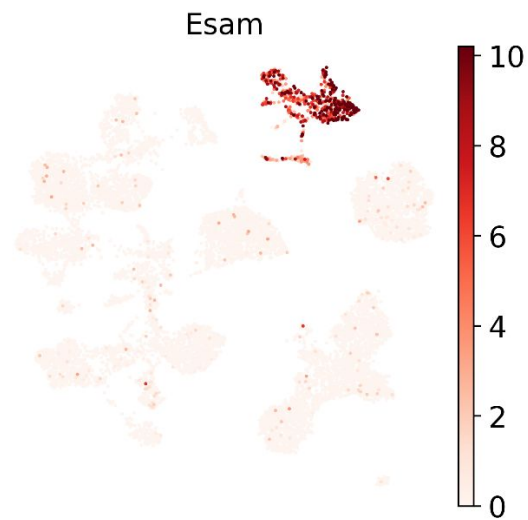
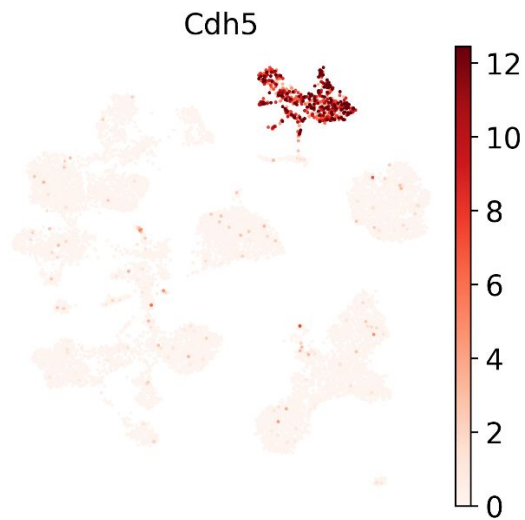
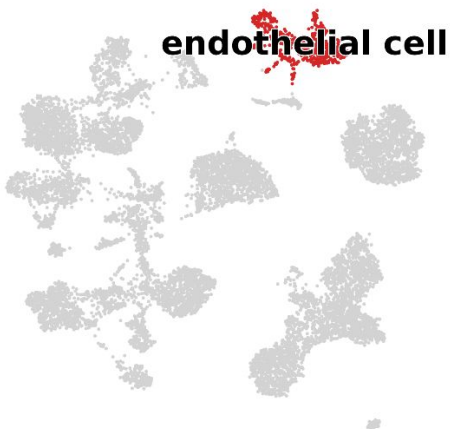
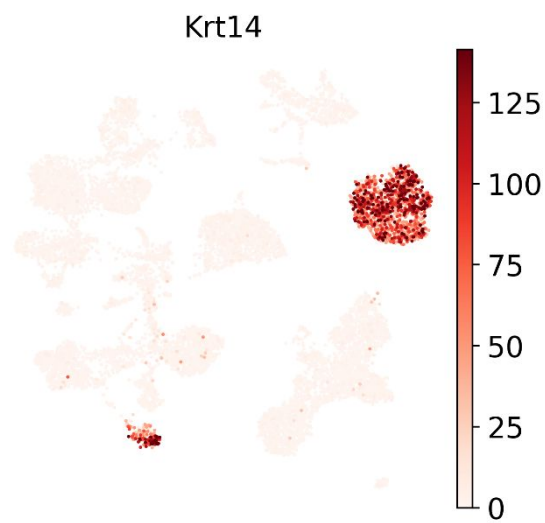
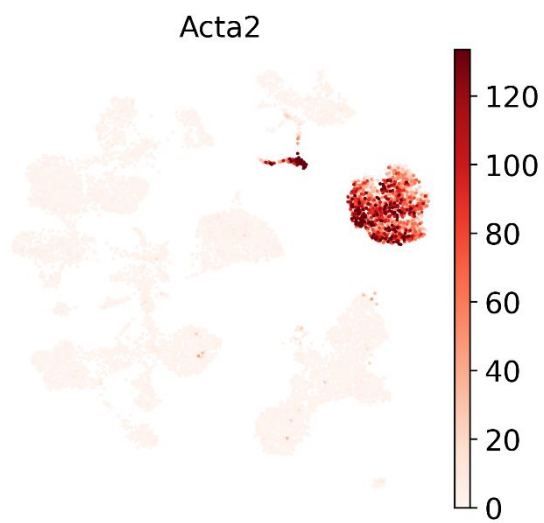
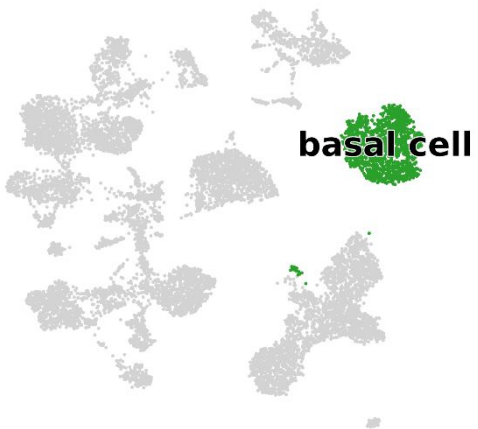
NSForest in mammary gland from Tabula Muris Senis

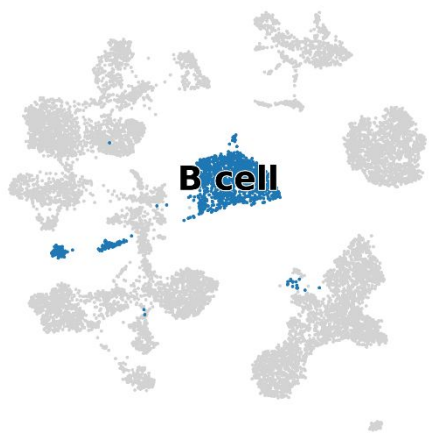


7 cell types

B cell	Cd79a, H2-Eb1
T cell	Cd3d, Cd3g
Basal cell	Acta2, Krt14
Endothelial cell	Cdh5, Esam
Luminal epithelial cell	Krt19, Cldn3
Macrophage	Tyrobp, Cd68
Stromal cell	Col1a2

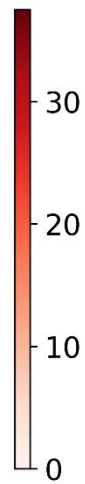
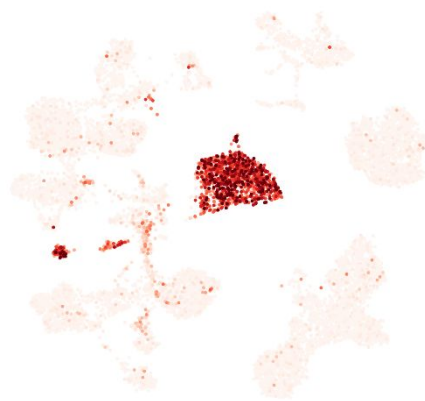
Total: 13 genes



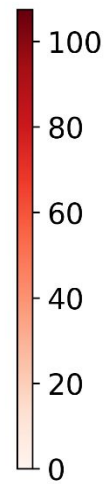
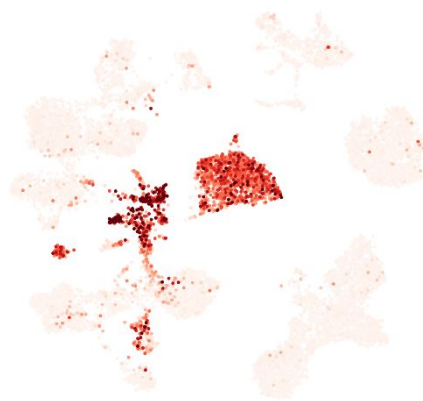


B cell

Cd79a

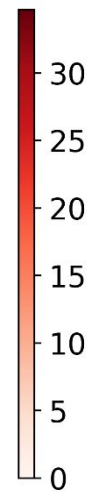
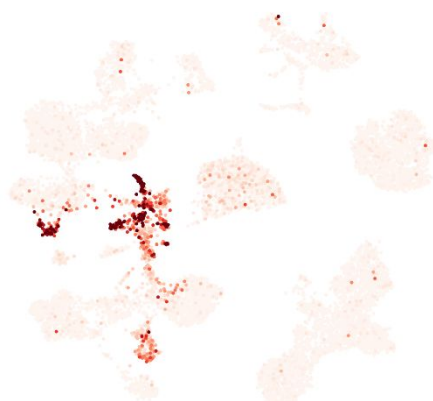


H2-Eb1

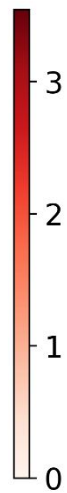


macrophage

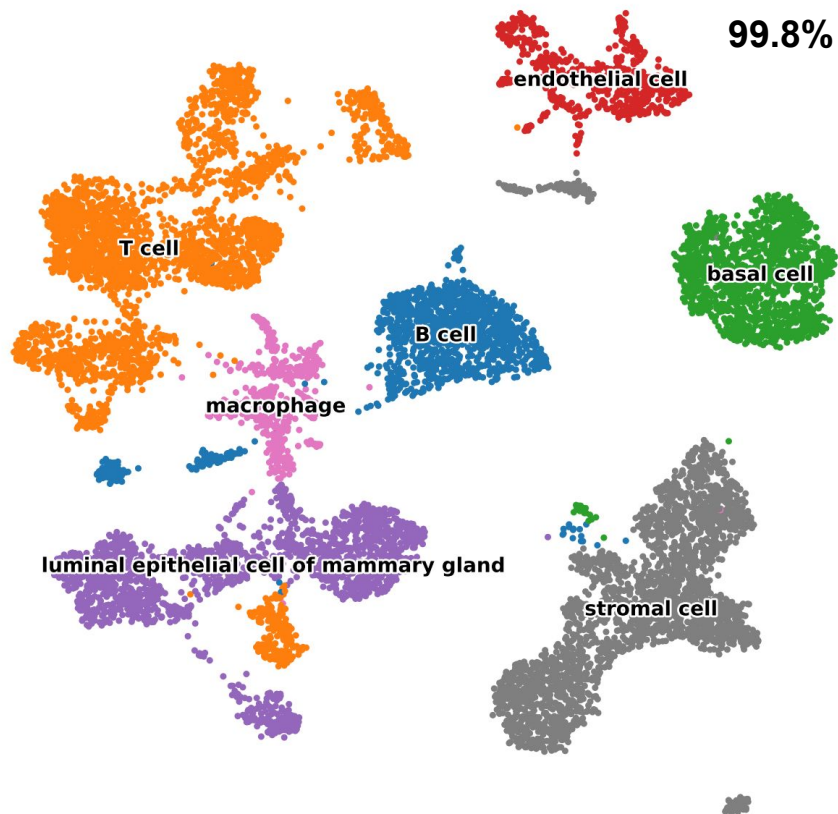
Tyrobp



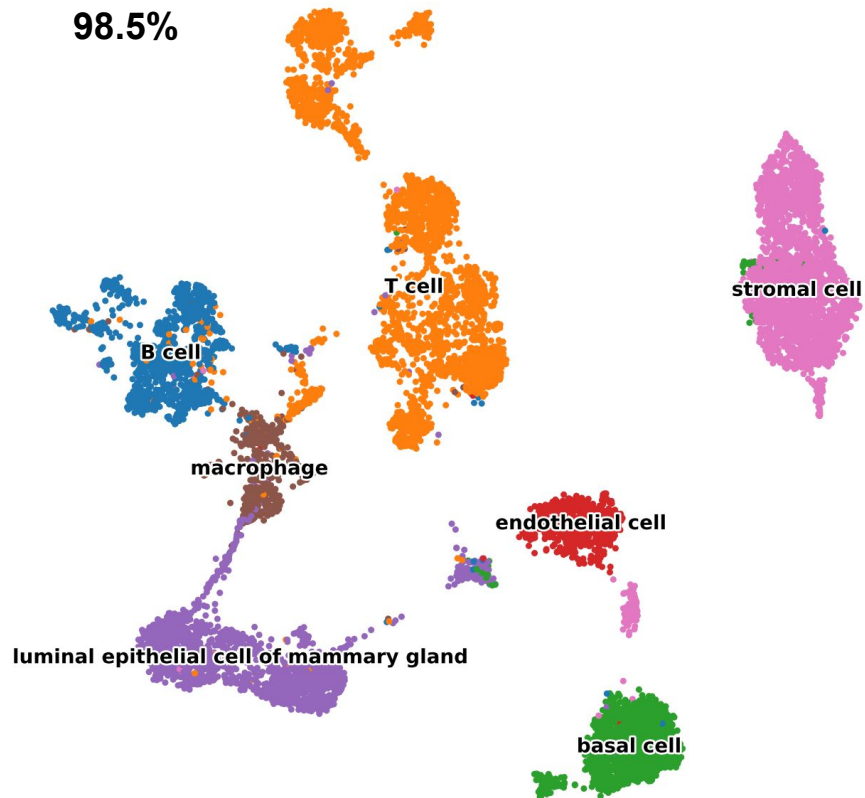
Cd68



Mammary gland cell types



MERFISH re-clustering



Codebook and Probe Design

MERFISH – Considerations for codebook design

- The “error robust” part of MERFISH comes from the codebook design
- In theory, an N-bit codebook can encode 2^N genes, but errors would cause misidentification of targets
 - Gene A: 110110010110
 - Gene B: 100110010110
- Too many genes targeted in one image can cause optical crowding, meaning light from nearby molecules bleed together preventing them from being identified as separate molecules
- Genes with different “weights”, e.g. 0000010000 and 1110111111, would likely have very different detection efficiency, introducing bias towards certain genes into the result.

MERFISH – Error robust codebook design

- Codebooks have consistent Hamming Weight (number of 1s)
 - Usually we use Hamming Weight 4 (HW4)
- Codes have a minimum Hamming Distance of 4
 - 00101000110000
 - 00001000110000 (Hamming distance 1)
 - 00001010110000 (Hamming distance 2)
 - 00001010010000 (Hamming distance 3)
 - 00001010010010 (Hamming distance 4)
- This enables correction of single bit errors and identification (and removal) of two bit errors. Three errors need to occur to have a misidentification.

MERFISH probe design

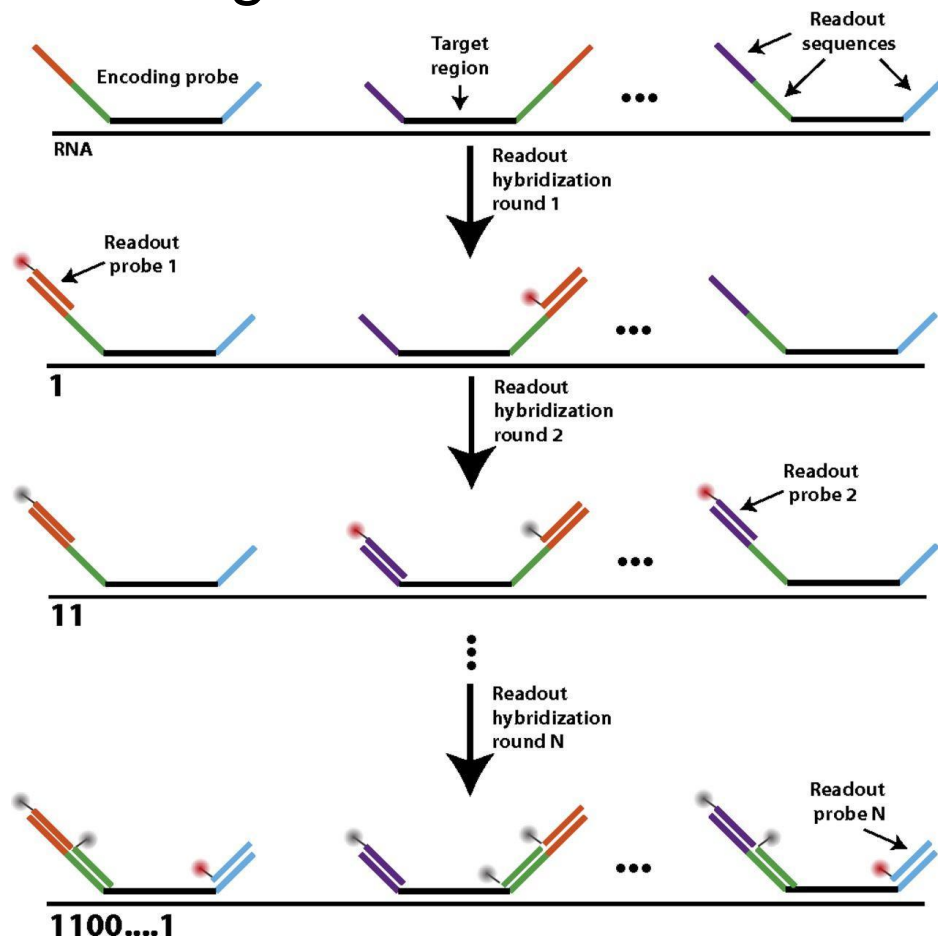
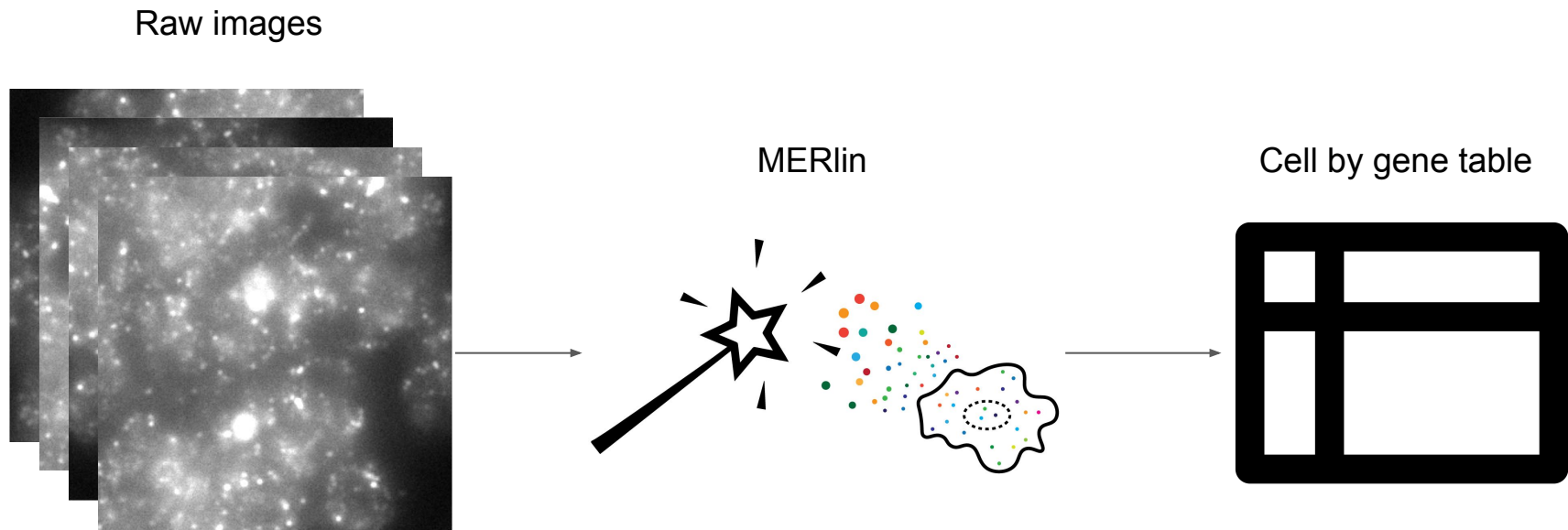
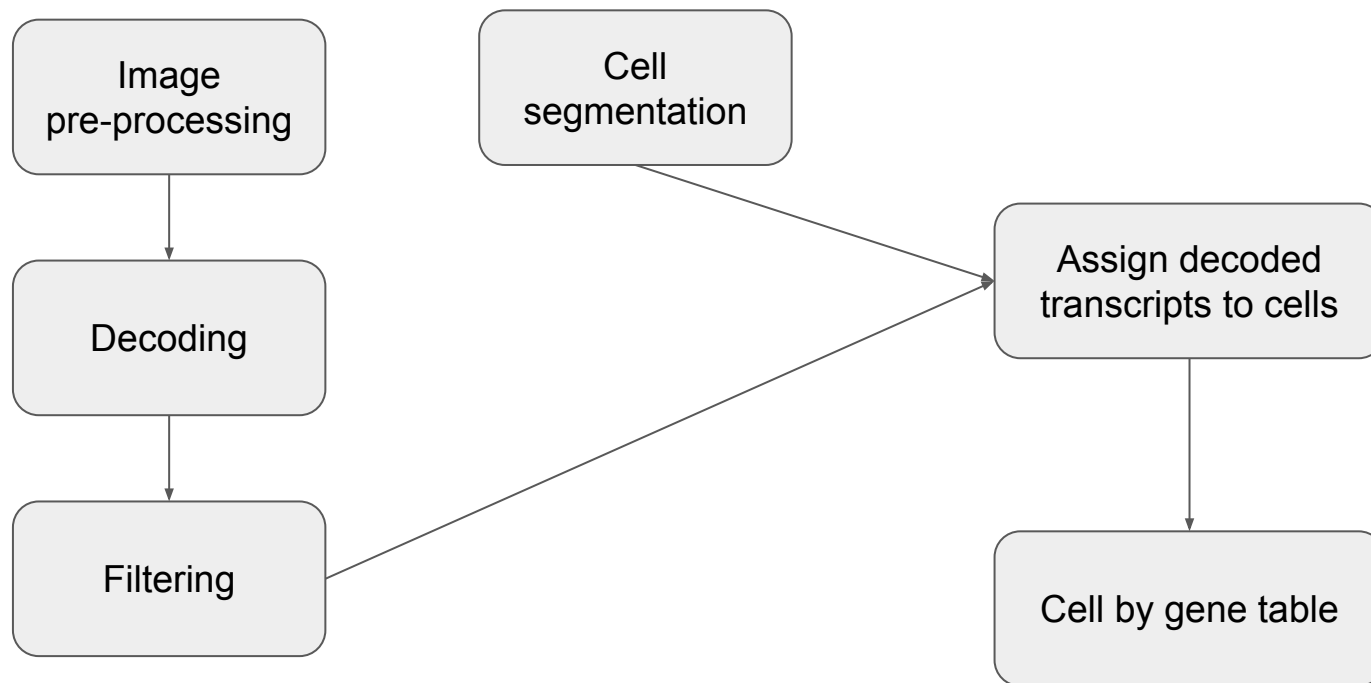


Image processing

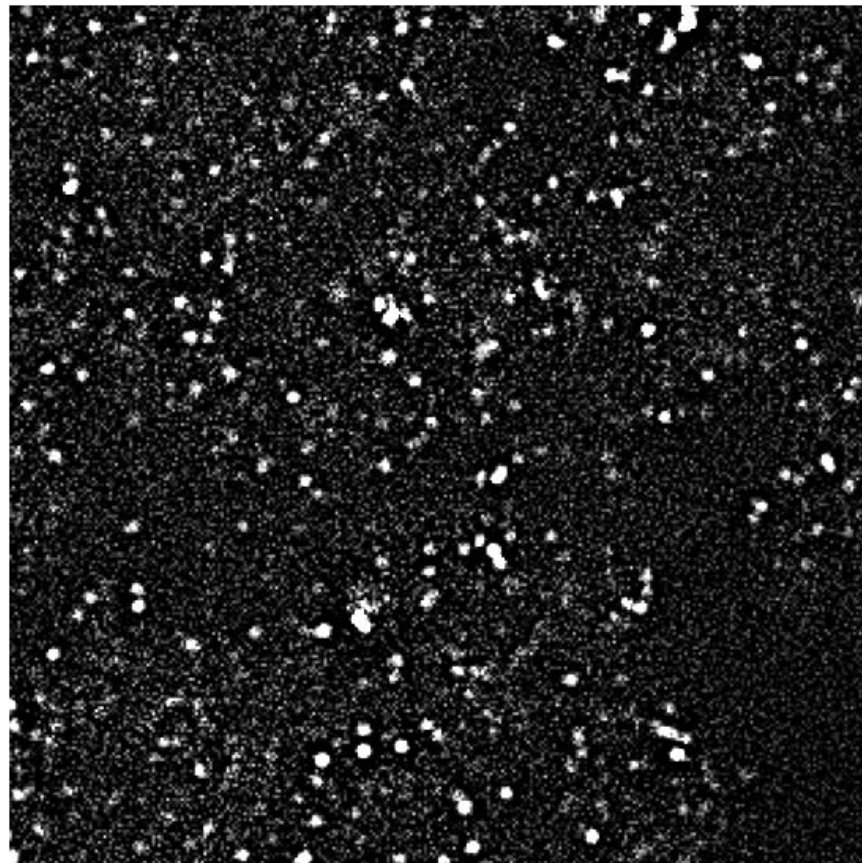
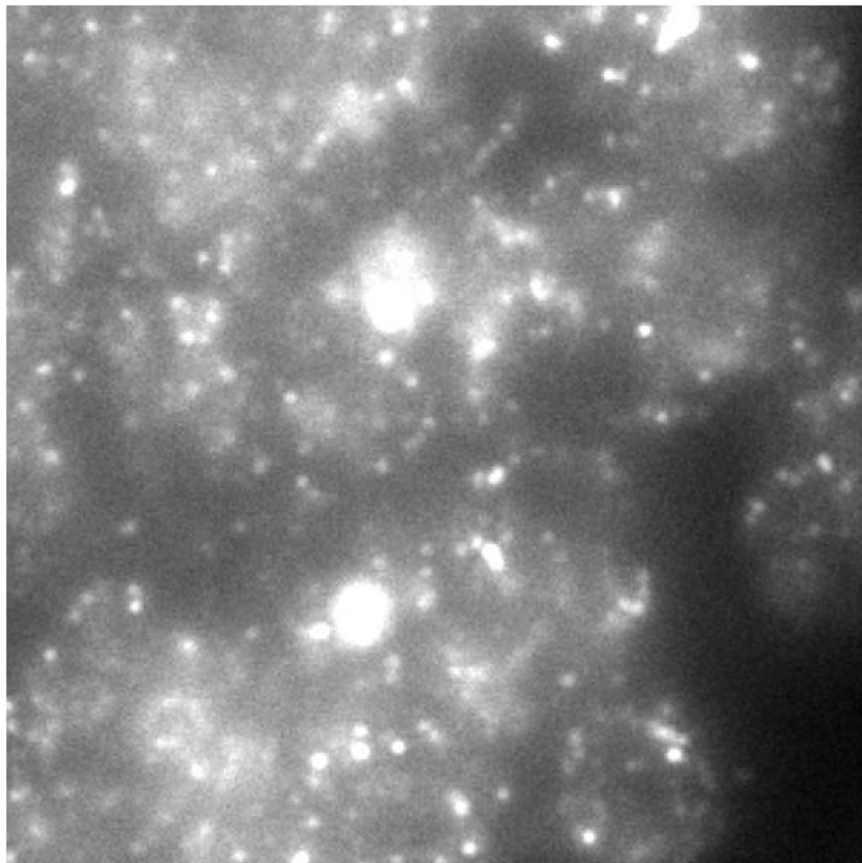
MERlin software developed in the Zhuang lab



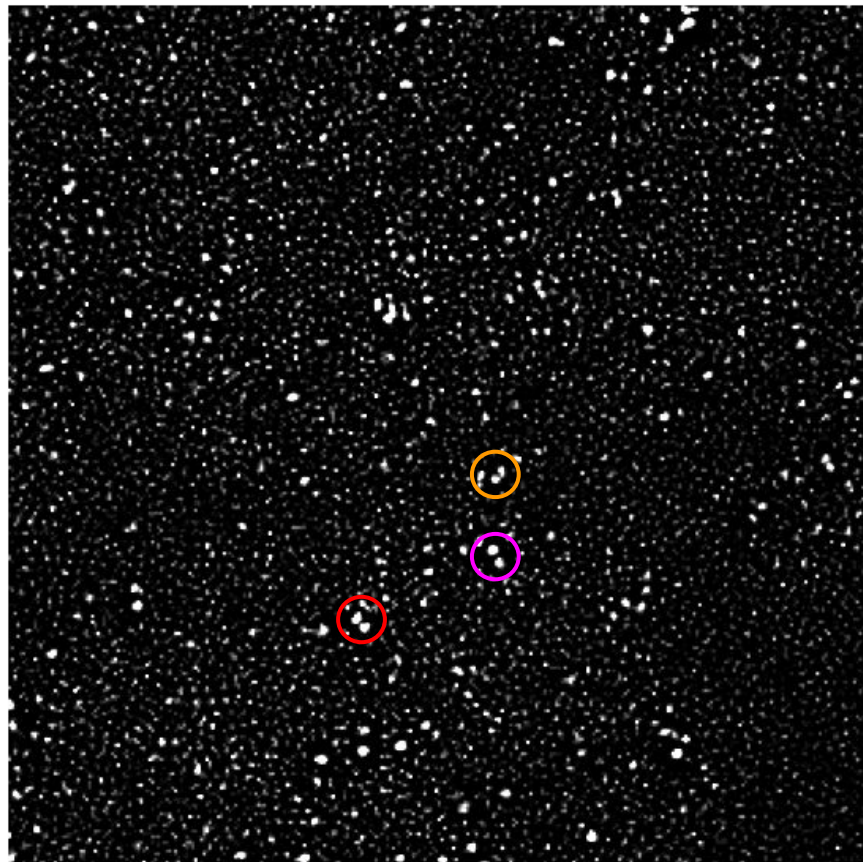
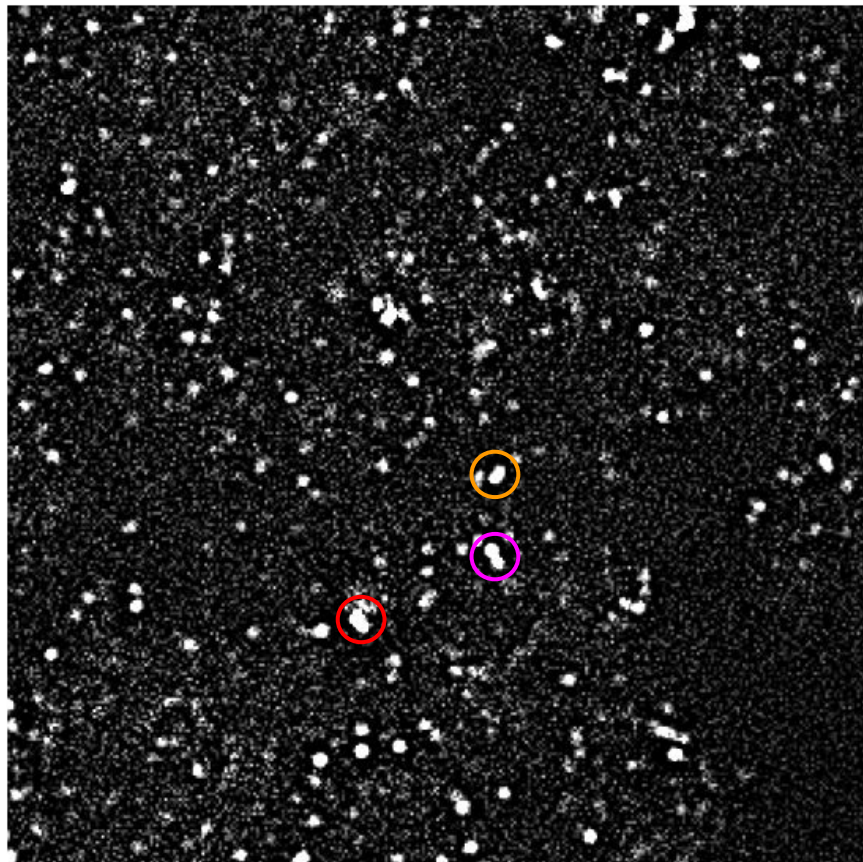
MERlin overview



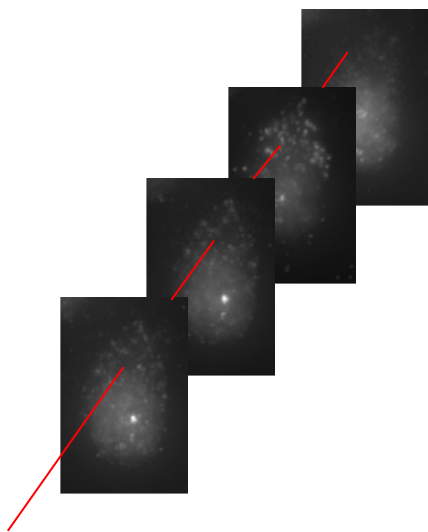
MERlin – High pass filter



MERlin – Deconvolution



MERlin – Pixel-based decoding



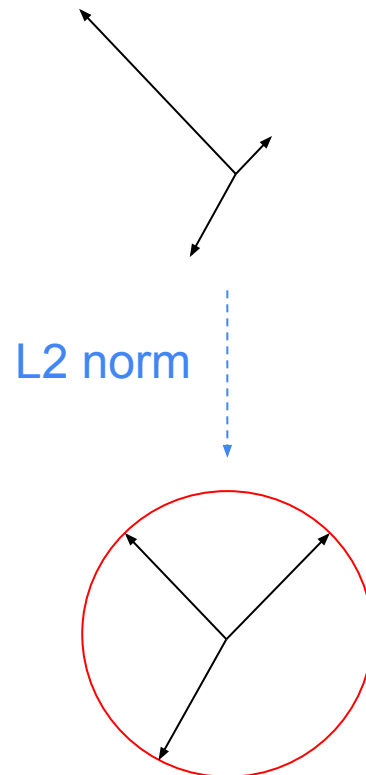
Pixel Traces	Bit 1	Bit 2	Bit 3	Bit 4	...	Bit N
Pixel 1	564	1237	146	678	...	1743
Pixel 2	404	768	449	1004	...	1344
Pixel 3	910	128	1874	2139	...	532
...	...	...	...	...	...	...
Pixel M	2342	1432	873	230	...	127

MERlin – Pixel-based decoding

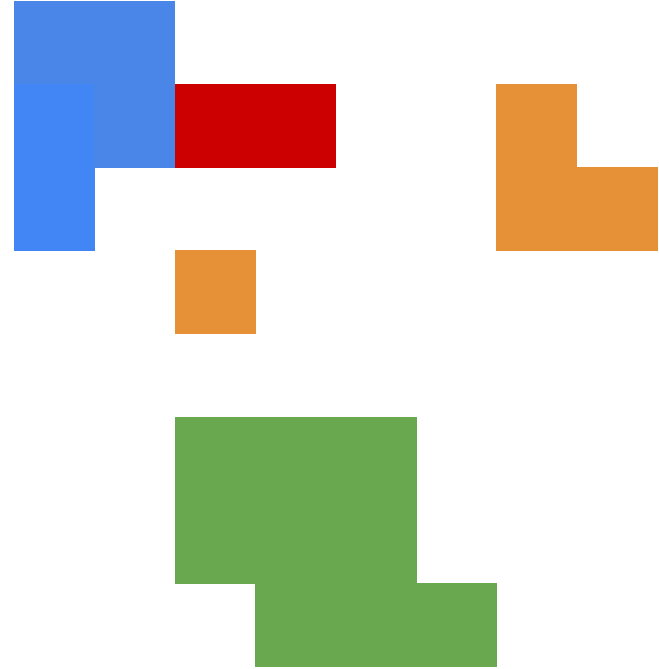
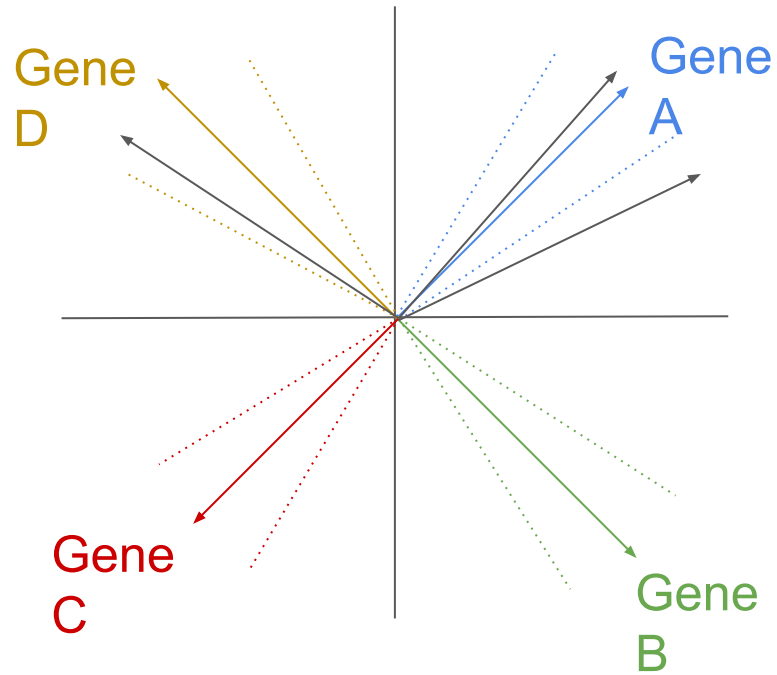
Pixel 1	564	1237	146	678	...	1743
Pixel 2	404	768	449	1004	...	1344
Pixel 3	910	128	1874	2139	...	532



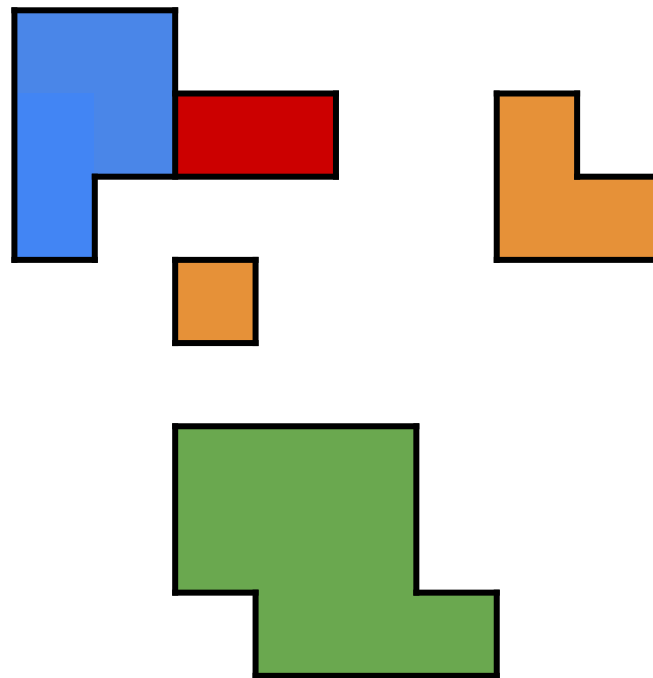
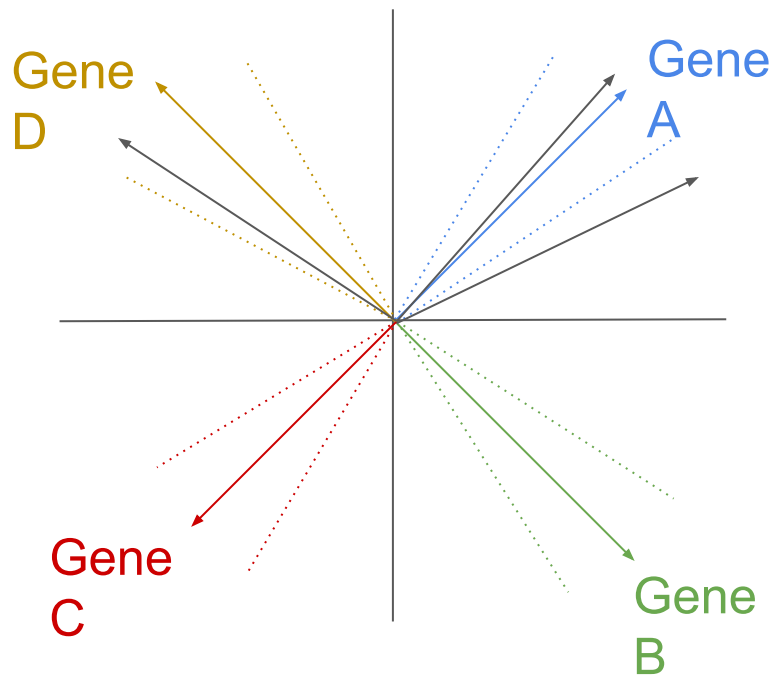
Pixel 1	0.243	0.534	0.063	0.293	...	0.752
Pixel 2	0.208	0.396	0.231	0.517	...	0.692
Pixel 3	0.300	0.042	0.617	0.704	...	0.175



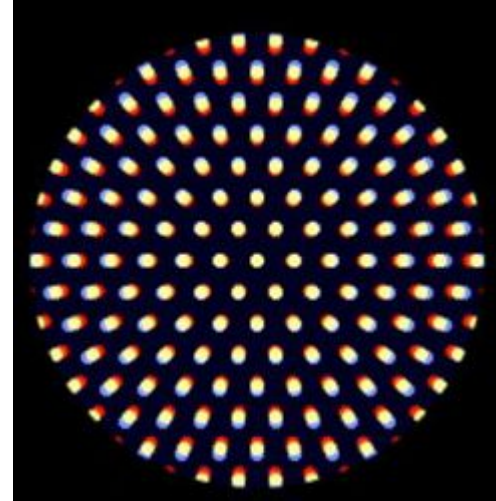
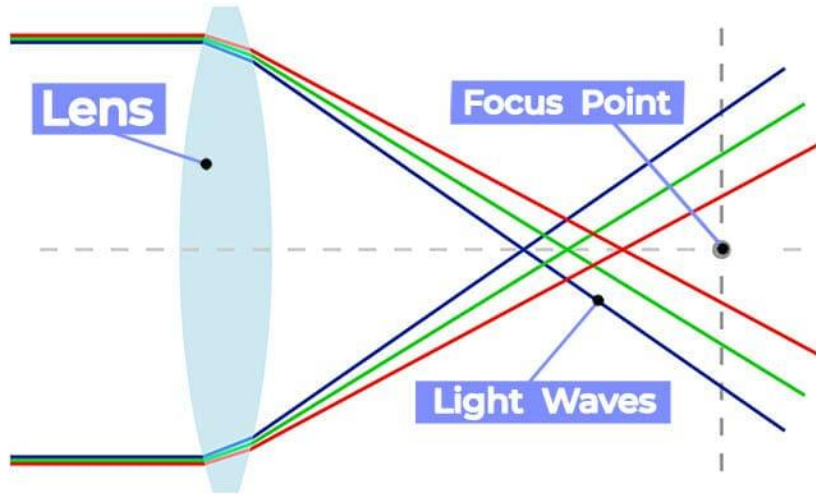
MERlin – Pixel-based decoding



MERlin – Pixel-based decoding

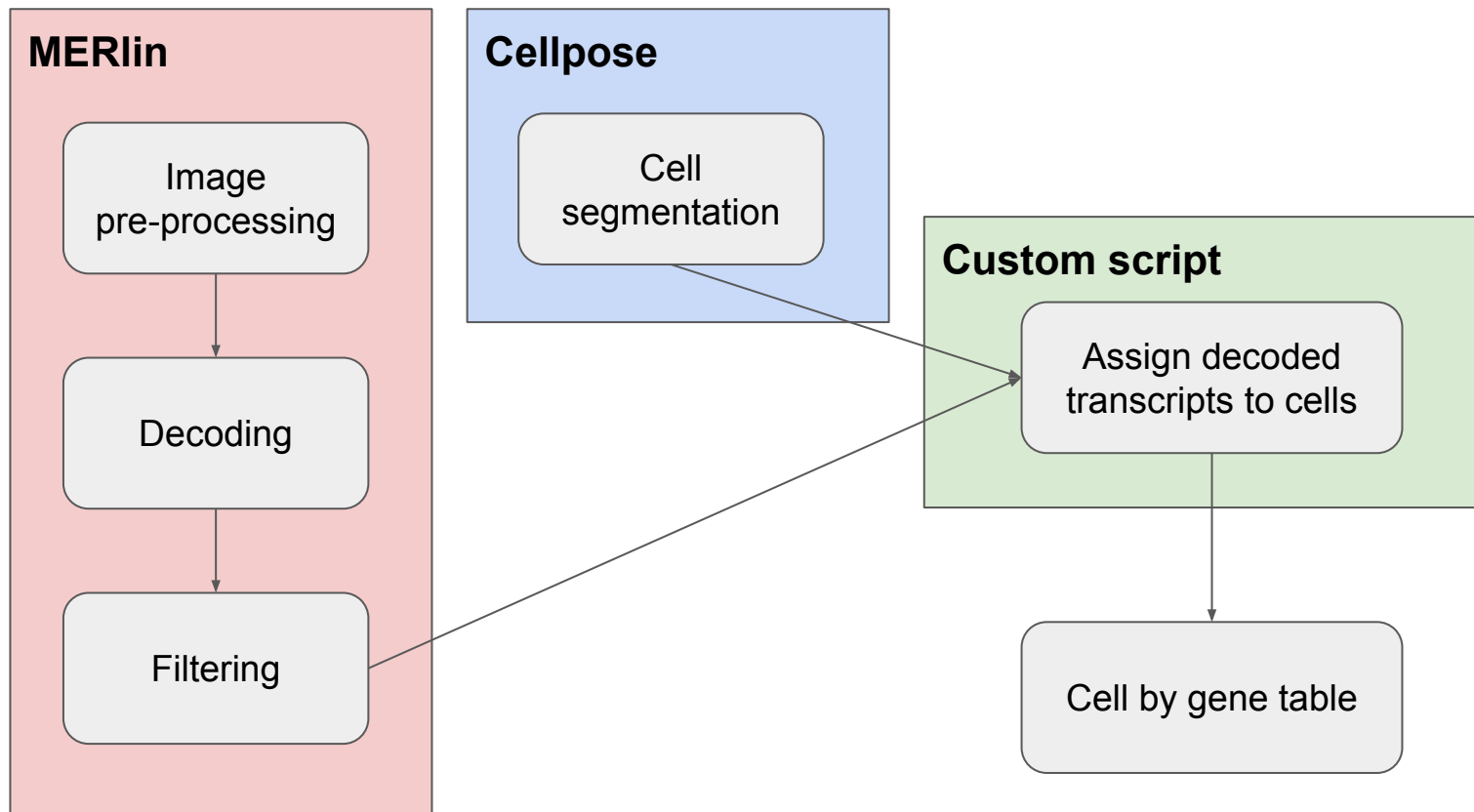


MERlin – Chromatic aberration correction



MERlin performs chromatic aberration correction at the same as brightness normalization by calculating the offset of different colored bits in each barcode.

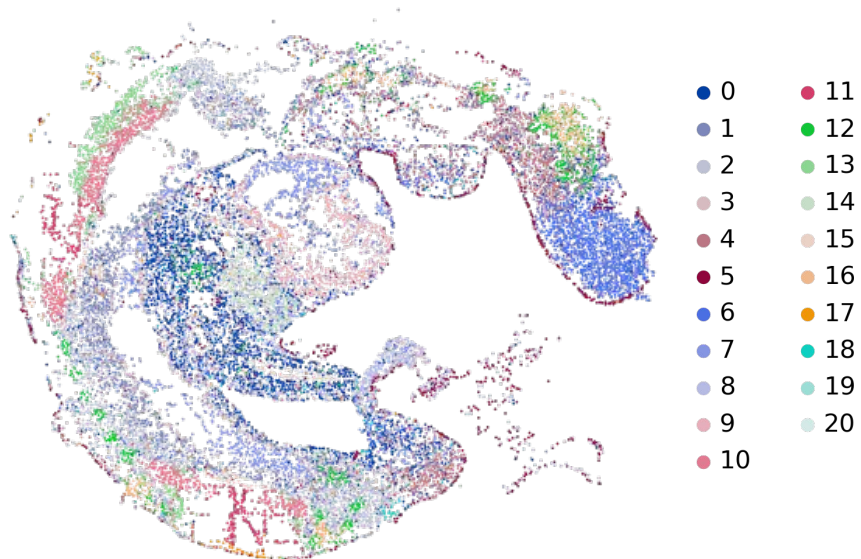
Established in-house MERFISH analysis pipeline



Cellpose segmentation improves MERSCOPE result

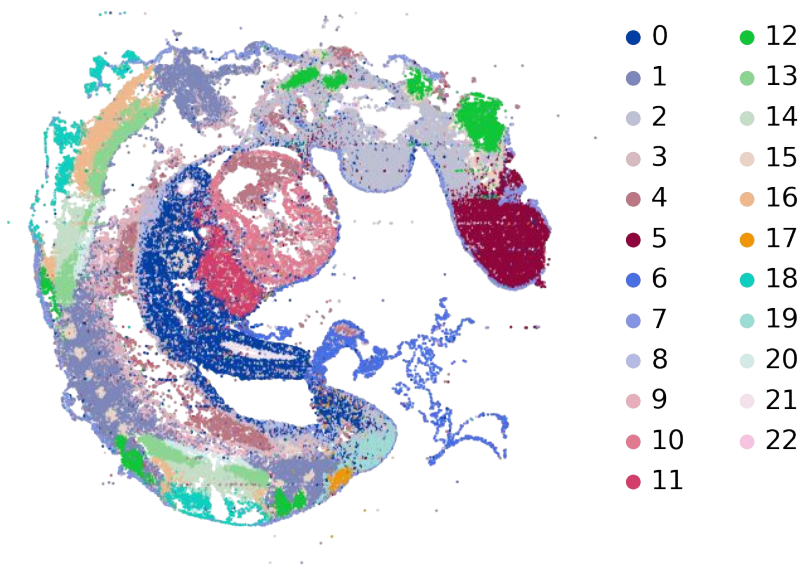
MERSCOPE segmentation

26,862 cells



Cellpose segmentation

61,429 cells



MERlin

Image pre-processing

Decoding

Filtering

Cellpose

Cell segmentation

Scanpy

Cell clustering

Custom scripts

Assign transcripts
to cells

QC metrics

Create cell by
gene table

Virtual FISH

Cell-to-cell
mapping

